通信与网络

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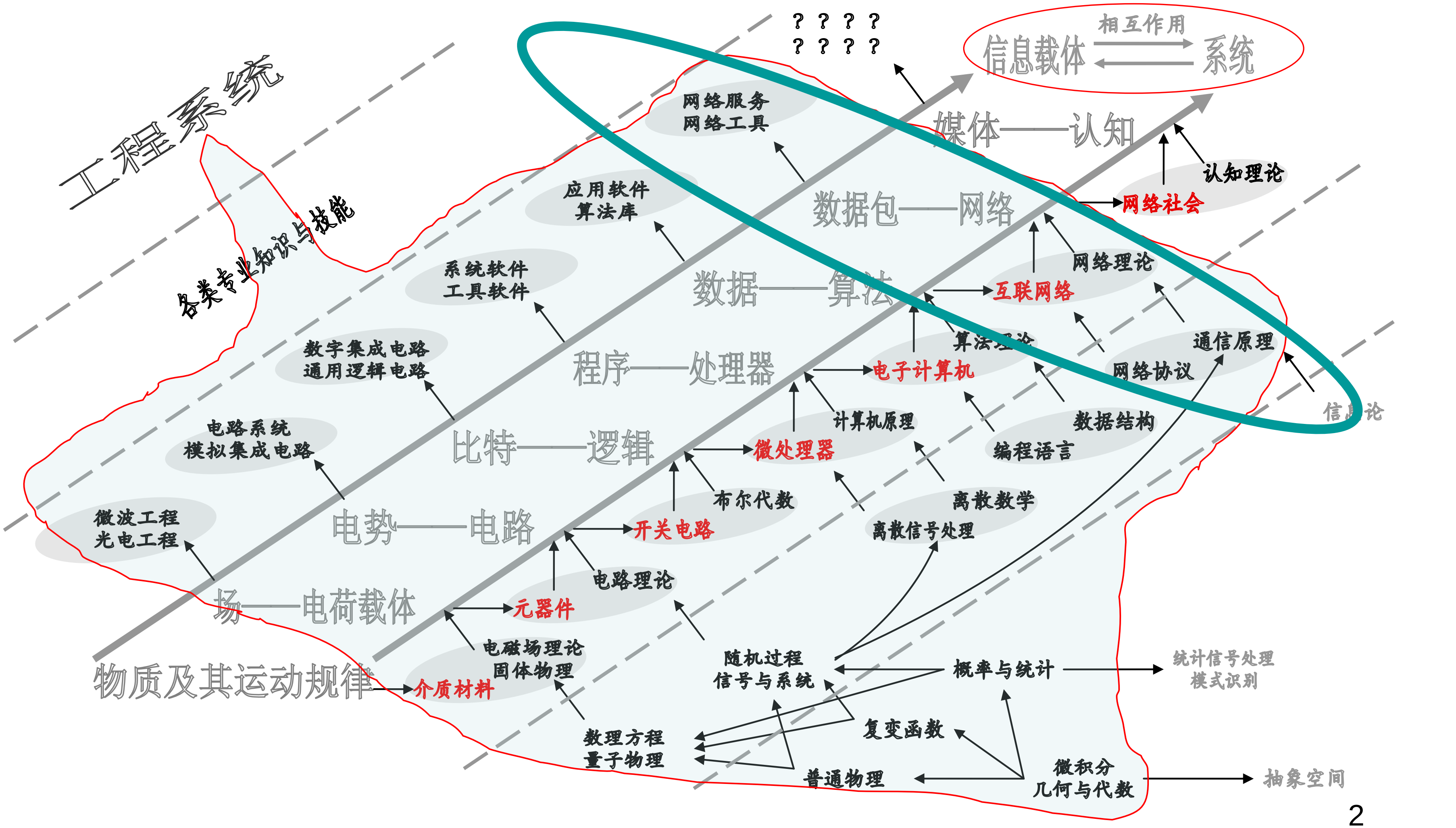
电子信息科学与技术知识体系



纲要

- 通信与网络概述
- 通信系统
- 通信理论
- 多址方式
- 网络协议
- 网络实例

通信与网络概述

基本概念

- 库仑定律、高斯定律、法拉第电磁感应定律、麦克斯韦-安培定律

$$
\begin{cases}
\oiint_S \boldsymbol{D} \cdot d\boldsymbol{S} = q \\
\oiint_S \boldsymbol{B} \cdot d\boldsymbol{S} = 0 \\
\oint_L \boldsymbol{E} \cdot d\boldsymbol{l} = -\iint_S \dfrac{\partial \boldsymbol{B}}{\partial t} \cdot d\boldsymbol{S} \\
\oint_L \boldsymbol{H} \cdot d\boldsymbol{l} = I + \iint_S \dfrac{\partial \boldsymbol{D}}{\partial t} \cdot d\boldsymbol{S}
\end{cases}
\qquad
\begin{cases}
\nabla \cdot \boldsymbol{D} = \rho \\
\nabla \cdot \boldsymbol{B} = 0 \\
\nabla \times \boldsymbol{E} = -\dfrac{\partial \boldsymbol{B}}{\partial t} \\
\nabla \times \boldsymbol{H} = \boldsymbol{j} + \dfrac{\partial \boldsymbol{D}}{\partial t}
\end{cases}
$$

基本概念

PHILOSOPHICAL MAGAZINE
AND
JOURNAL OF SCIENCE.

[FOURTH SERIES.]

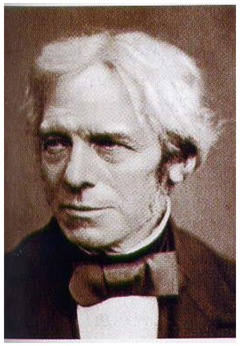
MARCH 1861.

XXV. *On Physical Lines of Force.* By J. C. MAXWELL, *Professor of Natural Philosophy in King's College, London**.

PART I.—*The Theory of Molecular Vortices applied to Magnetic Phenomena.*

IN all phenomena involving attractions or repulsions, or any forces depending on the relative position of bodies, we have to determine the *magnitude* and *direction* of the force which would act on a given body, if placed in a given position.

In the case of a body acted on by the gravitation of a sphere, this force is inversely as the square of the distance, and in a straight line to the centre of the sphere. In the case of two attracting spheres, or of a body not spherical, the magnitude and direction of the force vary according to more complicated laws. In electric and magnetic phenomena, the magnitude and direction of the resultant force at any point is the main subject of investigation. Suppose that the direction of the force at any point is known, then, if we draw a line so that in every part of its course it coincides in direction with the force at that point, this line may be called a *line of force*, since it indicates the direction of the force in every part of its course.



信息的内涵

- 信息是信息论中的一个术语，常常把消息中有意义的内容称为信息
  - 信息论的创始人香农在题为“通信的数学理论”论文中指出：“信息是用来消除随机不确定性的东西”
  - 熵的定义

$$H = -\sum_{n=1}^{N} p_n \log p_n$$

信息的内涵

- “前面有个人！”——非常含糊！

  需要1比特 —— “1”指人 “0”指其他

- “前面有个女人！”——具体！

  需要增加1比特 —— “1”指男人 “0”指女人

信息的内涵

- ➢“**信息是用来消除随机不定性的东西**”
- ➢信息和消息的不同

通信和网络

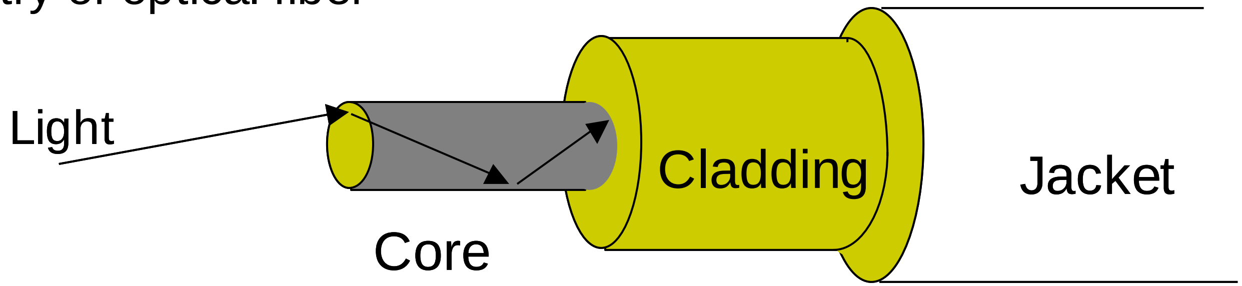
- 通信
  - 人类在生产和社会活动中总是伴随着信息的传递，这种信息的点到到点传递过程称为通信
  - 传输问题
- 网络
  - 网络，简单的来说，就是用物理链路将各个孤立的节点相连在一起
  - 交换问题

通信

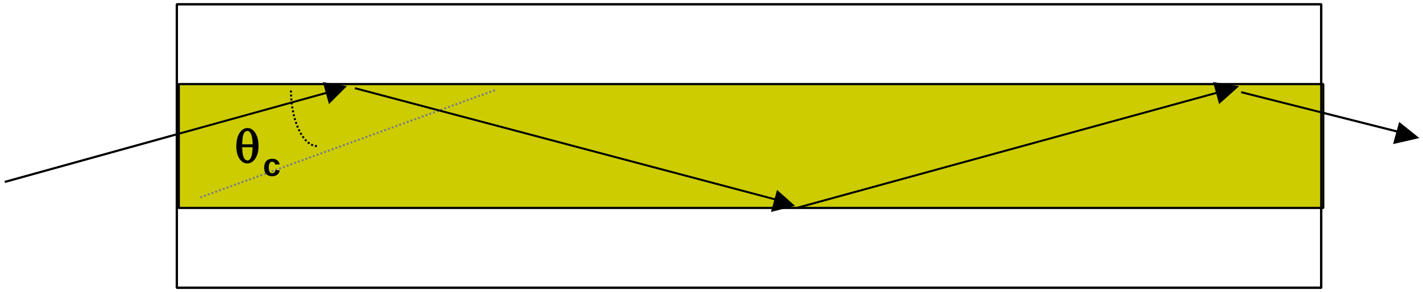
- 有线通信
  - 电缆
  - 光纤
  - 其他
- 无线通信
  - 电磁波
  - 可见光
  - 其他（量子等）
- 有关的课程
  - 随机过程、信号与系统、通信与网络、通信信号处理等

Transmission in Optical Fiber

Geometry of optical fiber

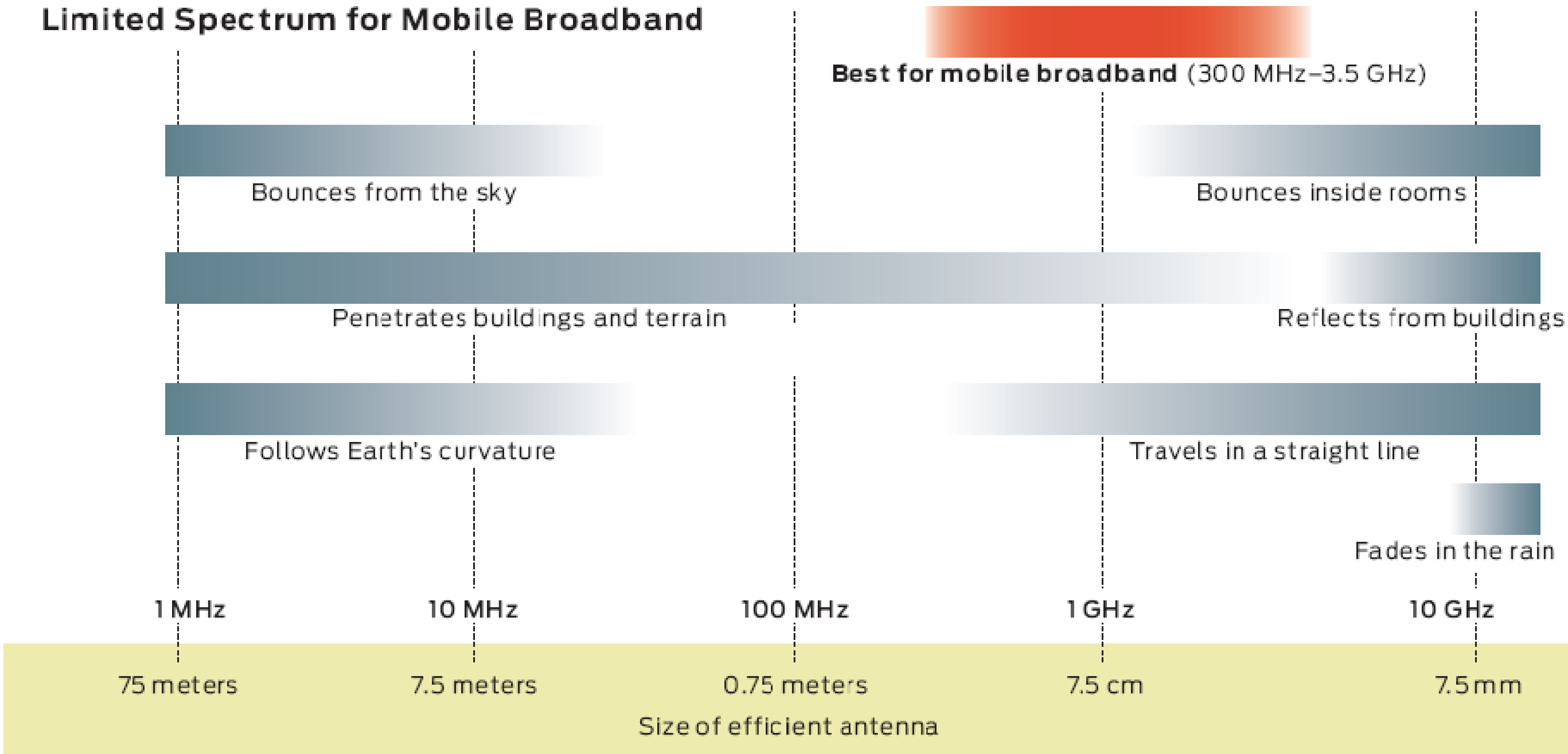


Total Internal Reflection in optical fiber



- Very fine glass cylindrical core surrounded by concentric layer of glass (cladding)
- Core has higher index of refraction than cladding
- Light rays incident at less than critical angle θ_c is completely reflected back into the core

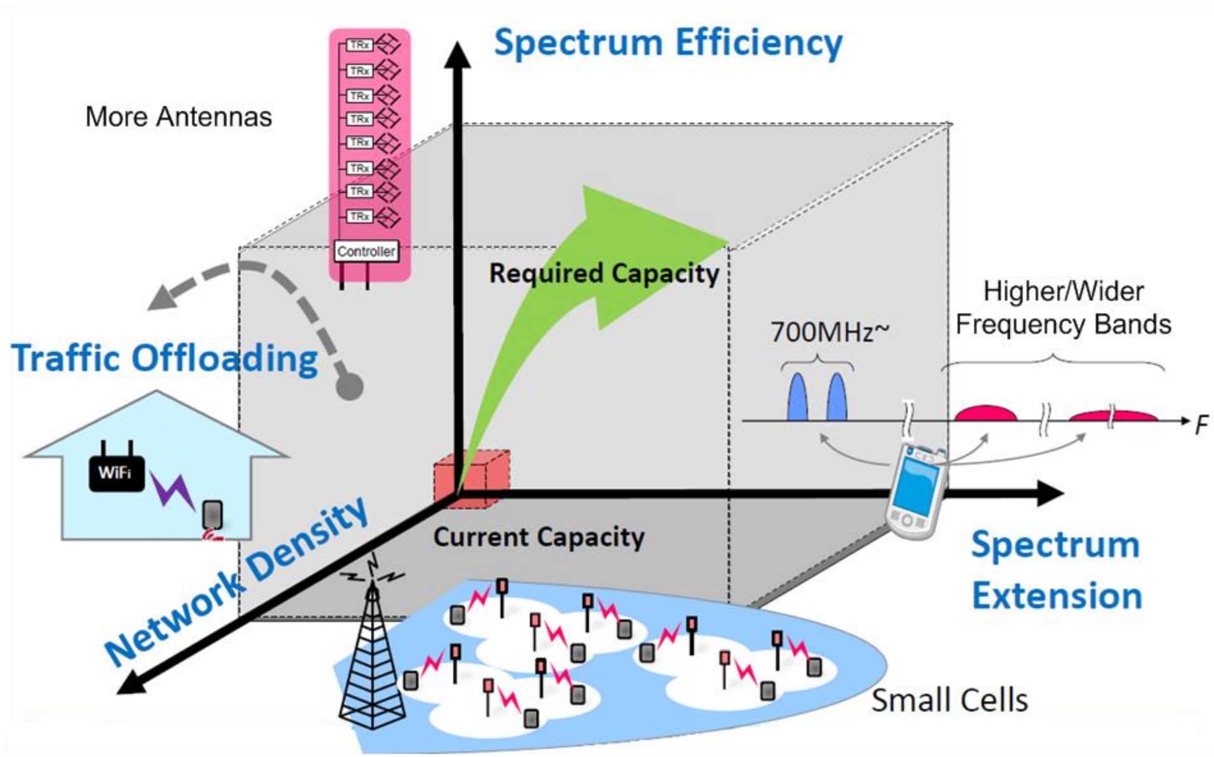
无线通信的频谱



OPPORTUNITY WINDOW: The best frequencies for mobile broadband are high enough that the antenna can be made conveniently compact, yet not so high that signals will fail to penetrate buildings. This leaves a relatively narrow range of frequencies available for use [red band].

无线通信的发展方向

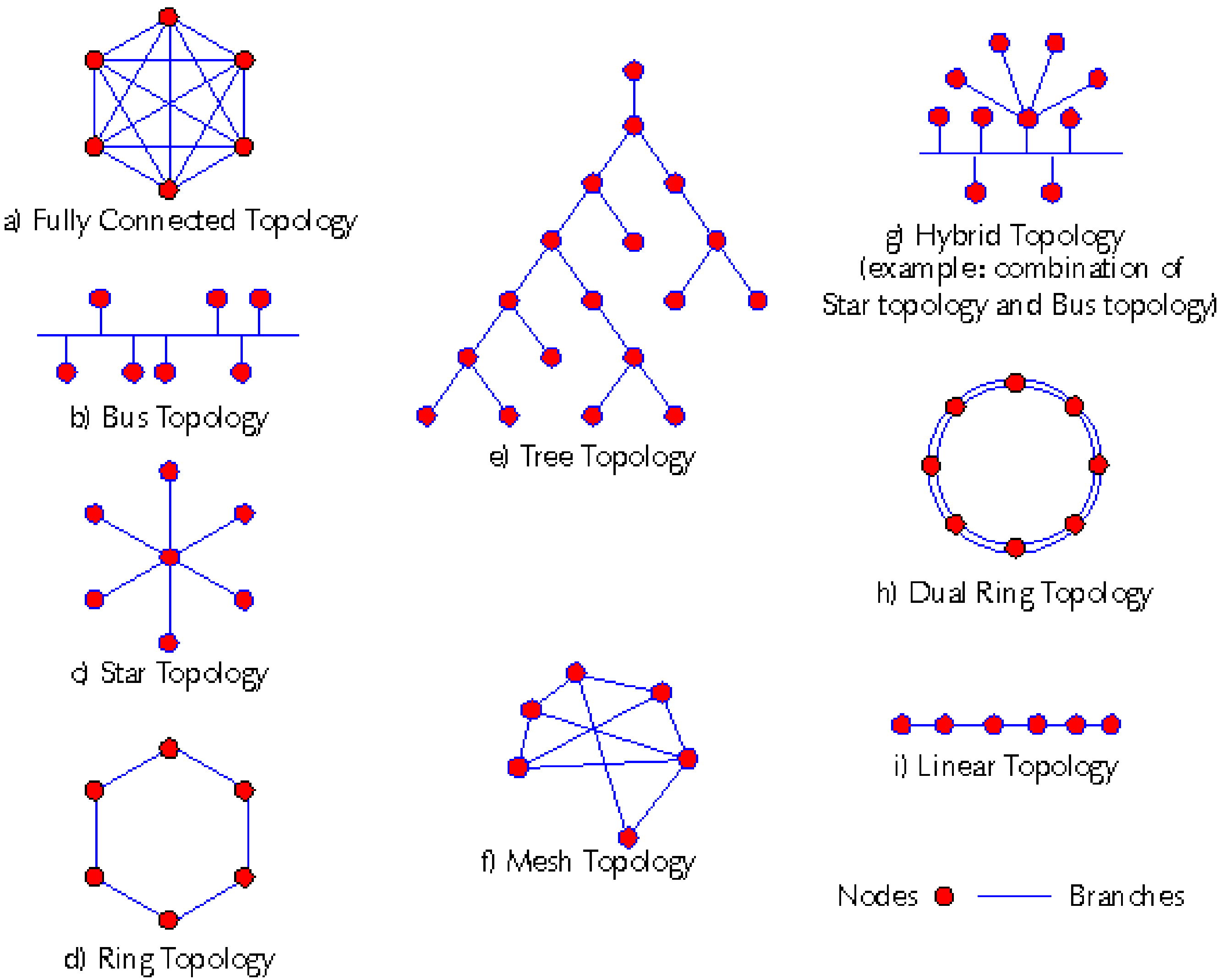
- 扩展频谱（同步、估计和均衡）
- 缩小蜂窝
- 多天线MIMO
- 融合WiFi



网络

- **网络**(Network)
  - **网络由节点集合与边集合组成，涵盖结构与行为两方面**
- **基于图论的形式化定义**
  - G(t) ={N(t); L(t); f (t) : J(t)}
  - N(t): nodes, also known as vertices or points
  - L(t): links, also known as edges or lines
  - f (t): N x N mapping function that connects node-pairs, yielding topology
  - **J(t): algorithm for describing behaviors of nodes and links versus time**

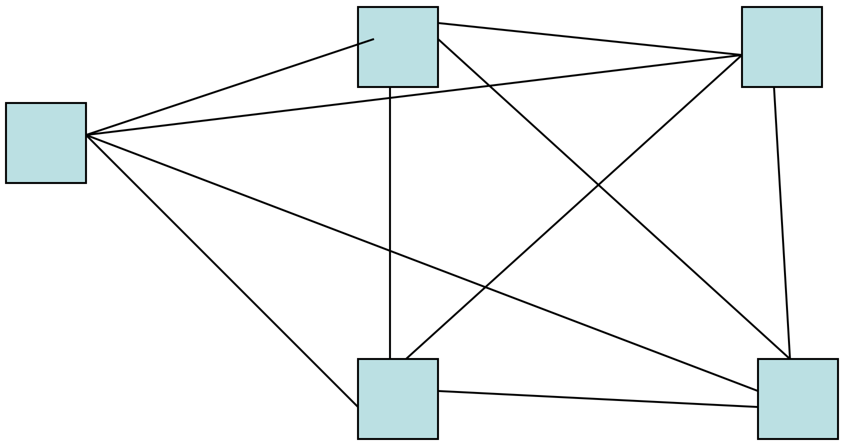
网络拓扑



a) Fully Connected Topology
b) Bus Topology
c) Star Topology
d) Ring Topology
e) Tree Topology
f) Mesh Topology
g) Hybrid Topology
(example: combination of
Star topology and Bus topology)
h) Dual Ring Topology
i) Linear Topology
Nodes
Branches

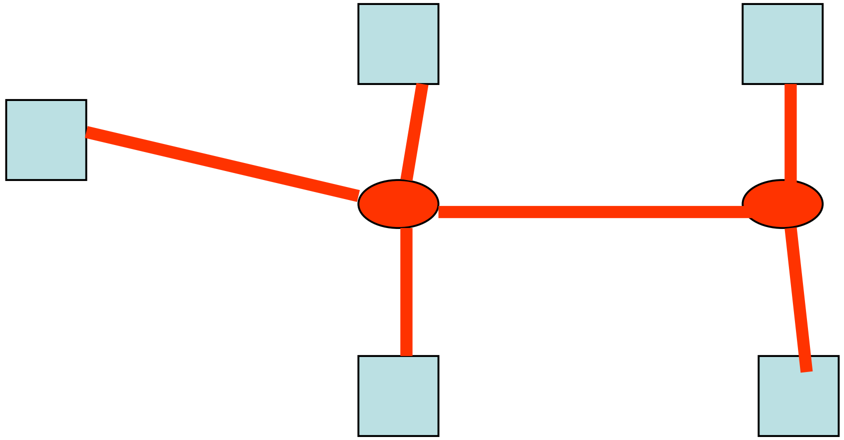
Evolution Networks （1）

- From point-to-point connection ...



Evolution Networks（2）

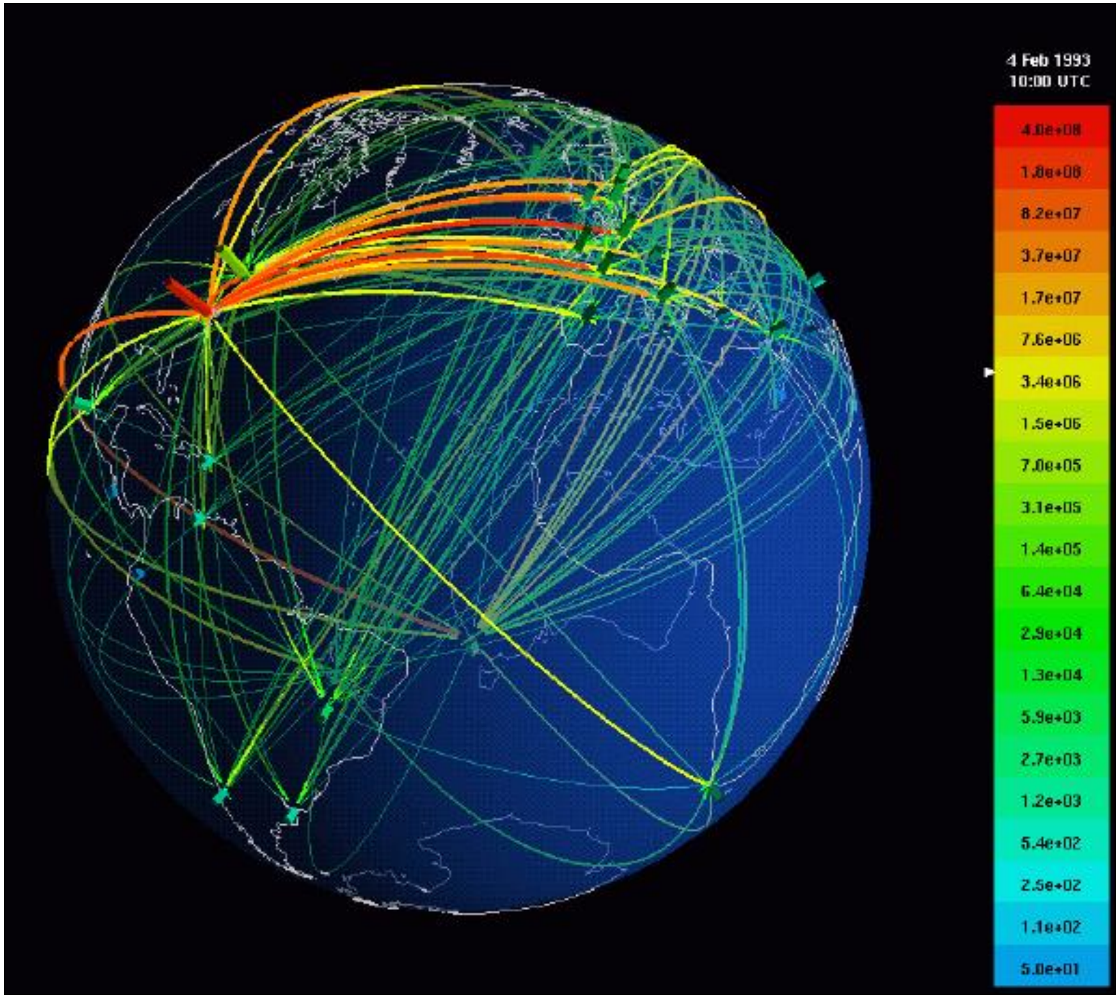
- **To shared links**



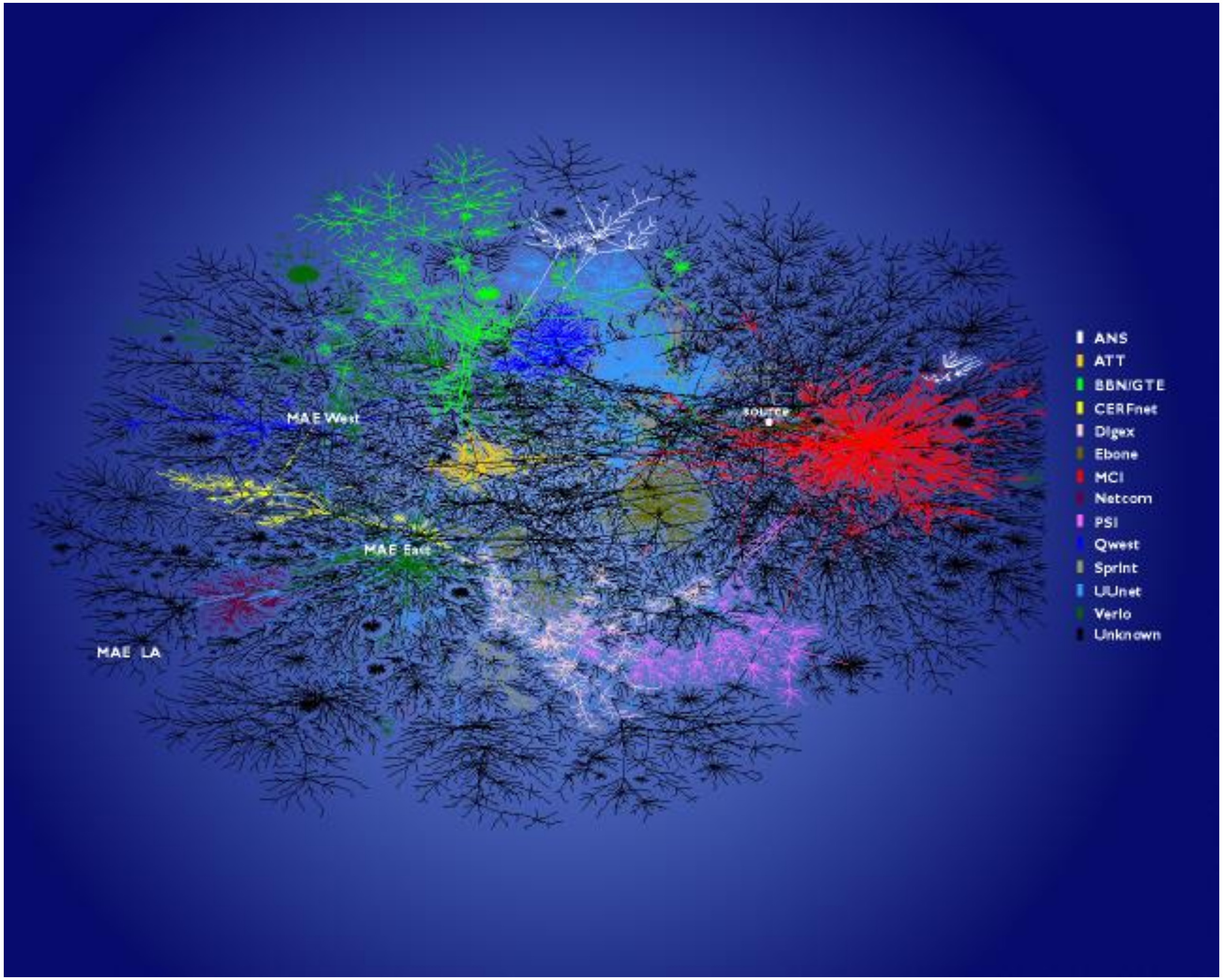
认识各种网络

- **简单规则网络**
  - 实验室小型局域网
- **复杂网络 —— 信息、技术(物理)、社会、生物**
  - 电信网络/无线网络
  - 万维网/HTTP
  - Internet
  - 生物网络（细胞网络、神经网络）
  - 人际网络
  - 科研合作网
  - …

电信网络

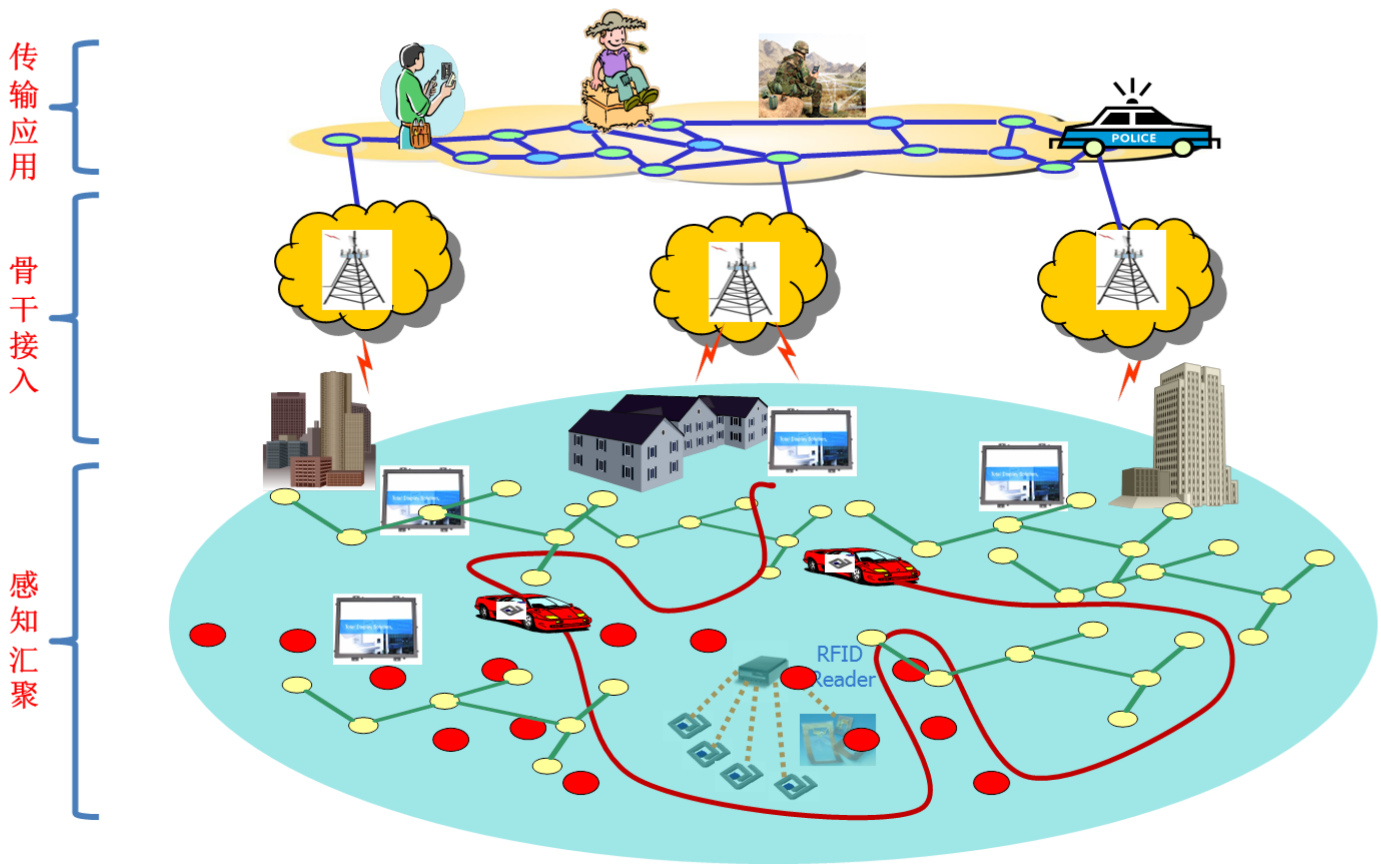

4 Feb 1993
10:00 UTC
4.0e+08
1.8e+08
8.2e+07
3.7e+07
1.7e+07
7.6e+06
3.4e+06
1.5e+06
7.0e+05
3.1e+05
1.4e+05
6.4e+04
2.9e+04
1.3e+04
5.9e+03
2.7e+03
1.2e+03
5.4e+02
2.5e+02
1.1e+02
5.0e+01

互联网



MAE West
ROUTER
MAE East
MAE LA
ANS
ATT
BBN/GTE
CERFnet
Digex
Ebone
MCI
Netcom
PSI
Qwest
Sprint
UUnet
Verio
Unknown

物联网



通信系统

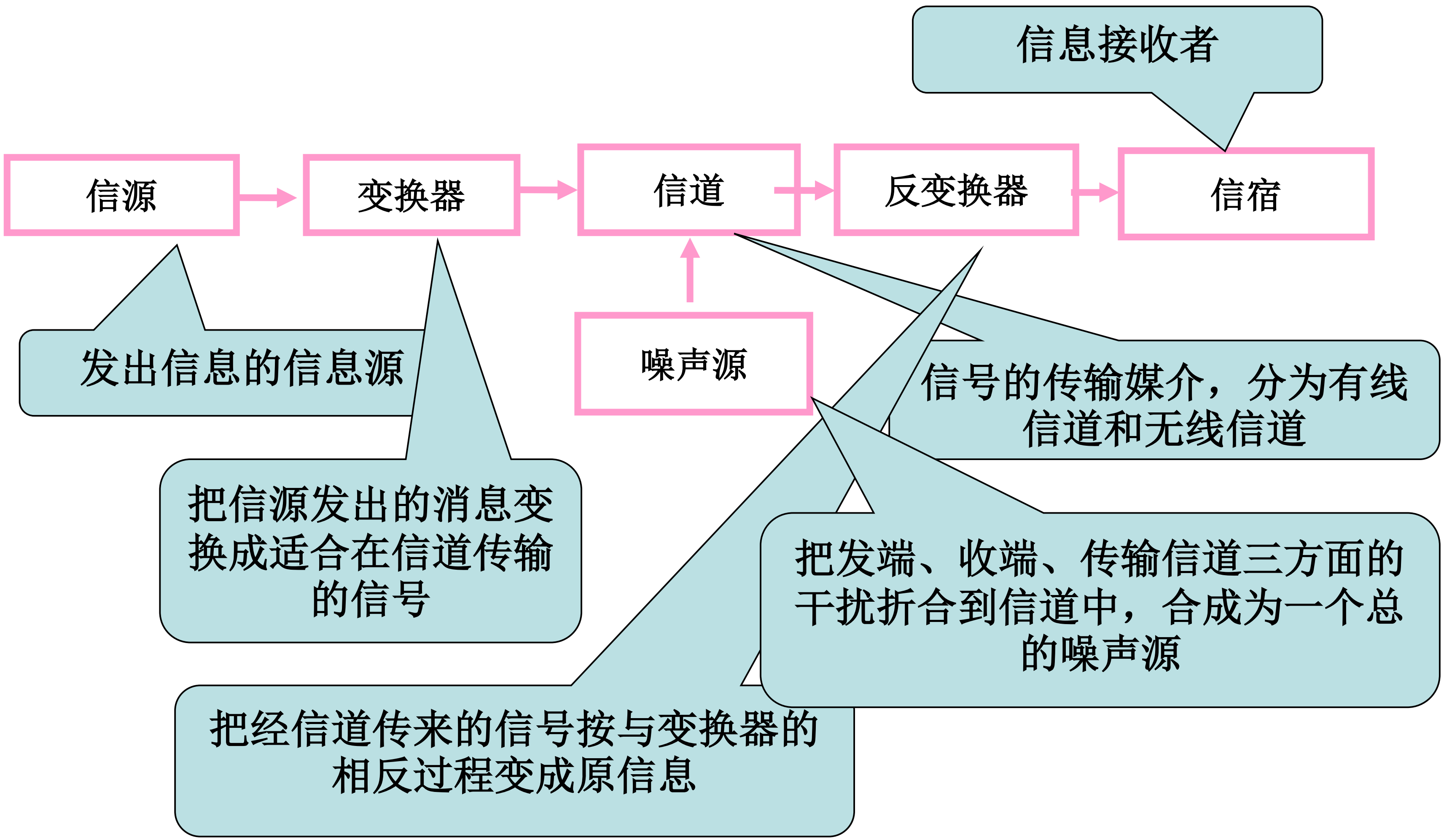
通信系统

- 模拟通信
  - 变换
  - 调制/解调
- 数字通信
  - 抽样
  - 贝叶斯和最大似然
  - 信息论和编码理论

通信技术发展概况

1838	莫尔斯 有线电报	1948	晶体管　香农信息论 统计理论
1864	麦克斯韦尔 电磁辐射方程	1950	时分多路通信 应用于电话
1876	贝尔 电话	1956	越洋电话铺设
1896	马可尼 无线电报	1957	第一颗人造卫星发射
1906	真空管	1958	第一颗通信卫星发射
1918	调幅广播 超外差接收机	1960	发明激光
1925	三路明线载波电话 多路通信	1961	发明集成电路
1936	调频广播	1962	第一颗同步通信卫星 PCM进入实用
1937	脉冲编码调制	1960	彩电 数字传输理论 高速计算机
1938	电视广播	1970	LSI 商用卫星 程控交换 光纤通讯
1940	二战 雷达和微波系统发展	1980	SLSI 长波光纤通信 ISDN 3G

通信系统模型

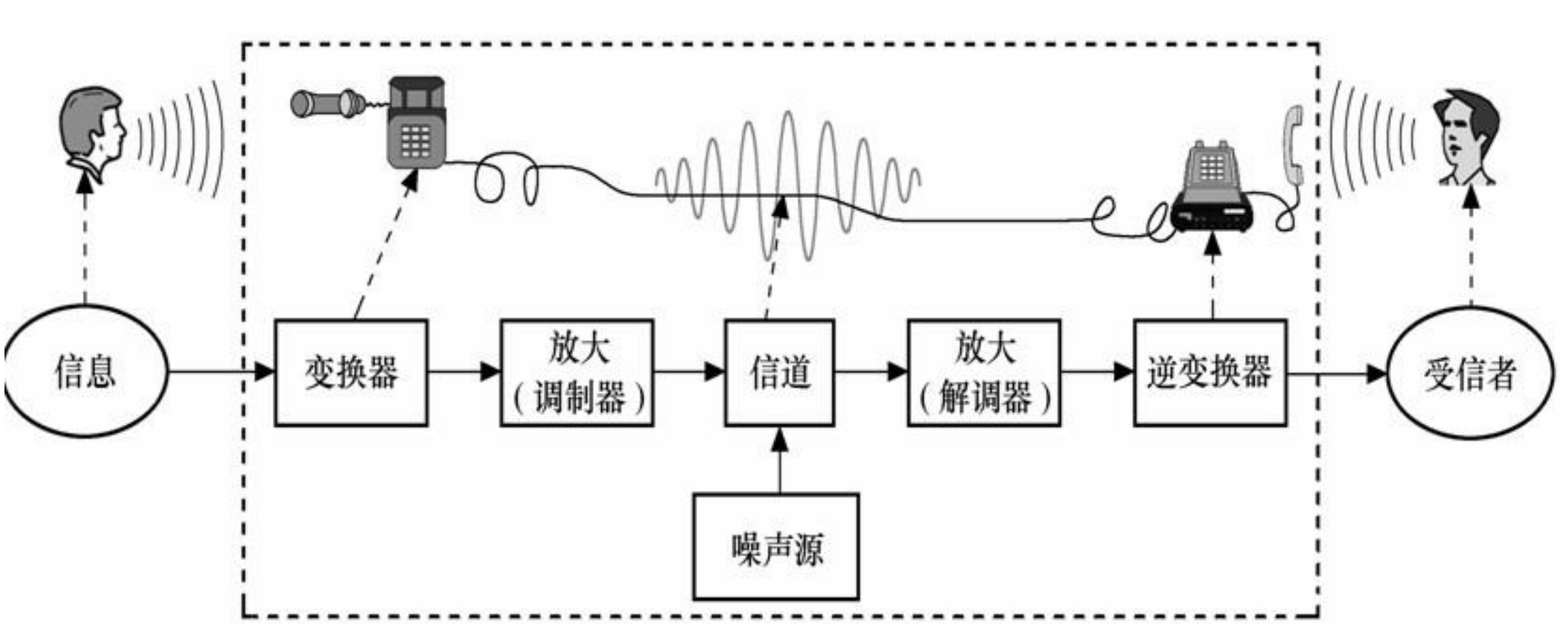

信息接收者
信源
变换器
信道
反变换器
信宿
噪声源
发出信息的信息源
把信源发出的消息变换成适合在信道传输的信号
信号的传输媒介，分为有线信道和无线信道
把发端、收端、传输信道三方面的干扰折合到信道中，合成为一个总的噪声源
把经信道传来的信号按与变换器的相反过程变成原信息

模拟通信系统模型

发信者 → 非电/电转换器 → 调制器 → 信道 → 解调器 → 电/非电转换器 → 信宿

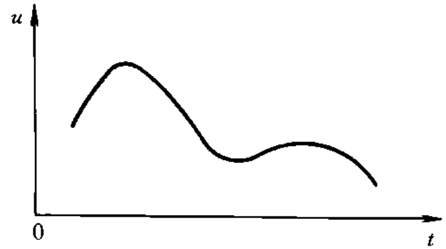
噪声源 → 信道

原始信号具有频率很低的频谱分量，不适合直接传输，需将其变换成其频带适合信道传输的信号

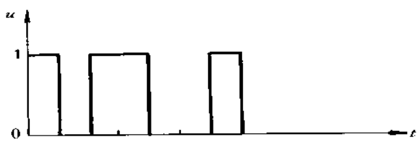


模拟通信和数字通信

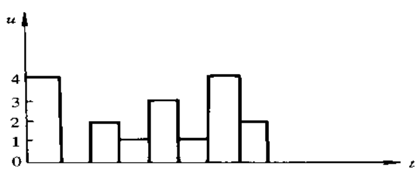
模拟信号：指代表信息的信号及其参数（幅度、频率或相位）随着信息连续变化的信号。



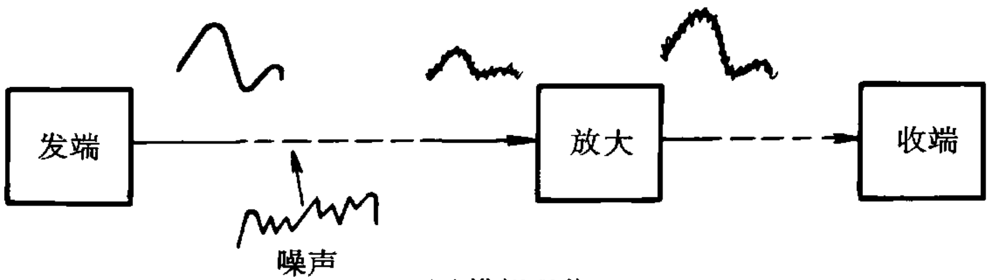
数字信号：不仅在时间上是离散的而且在幅度上也是离散的信号。



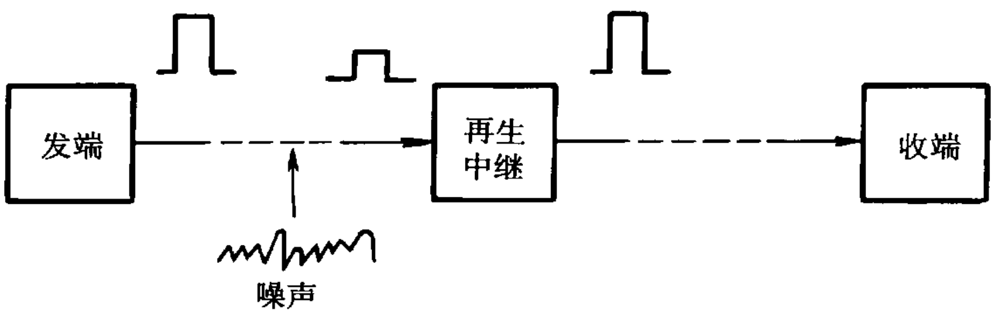
(a)



(b)

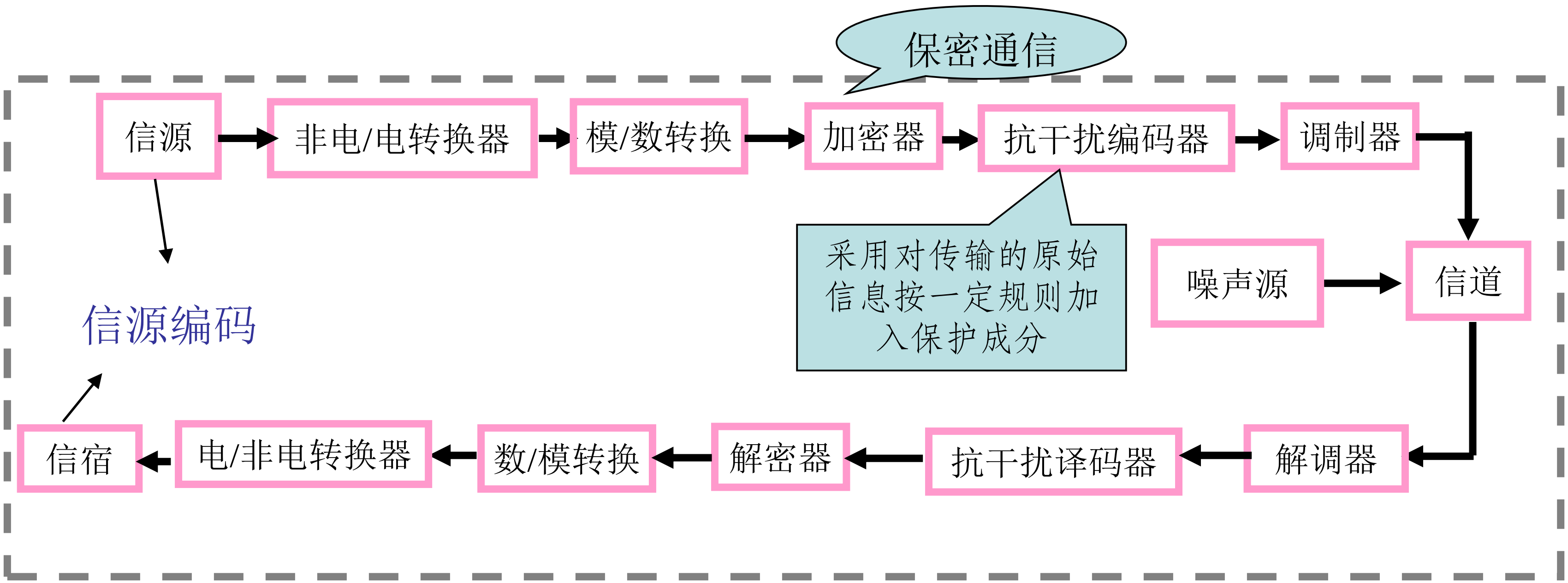


(a)模拟通信

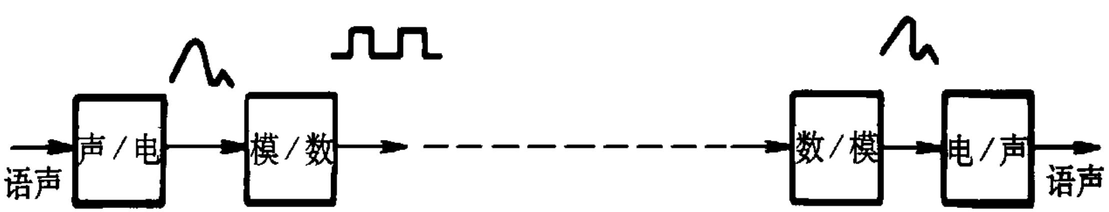


(b)数字通信

数字通信系统模型



保密通信
信源
非电/电转换器
模/数转换
加密器
抗干扰编码器
调制器
采用对传输的原始信息按一定规则加入保护成分
噪声源
信道
信源编码
信宿
电/非电转换器
数/模转换
解密器
抗干扰译码器
解调器



语声
声/电
模/数
数/模
电/声
语声

数字通信的特点和性能指标

特点

- **抗干扰能力强，无噪声积累**

  在数字通信中，由于数字信号的幅度值为有限个数的离散值(通常取两个幅值0、1)，在传输过程中受到噪声干扰虽然也要叠加噪声，但当信噪比还没有恶化到一定程度时，即在适当的距离，采用再生的方法即可消除噪声干扰，恢复原发送的信号。
- **灵活性强适应各种业务要求**
- **便于与计算机连接**
- **易于加密**

性能指标

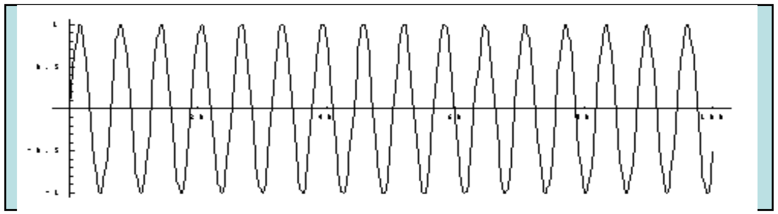
- **传输速率**

  信道的传输速率是以每秒钟所传输的信息量来衡量的，单位是比特每秒
- **频带利用率**

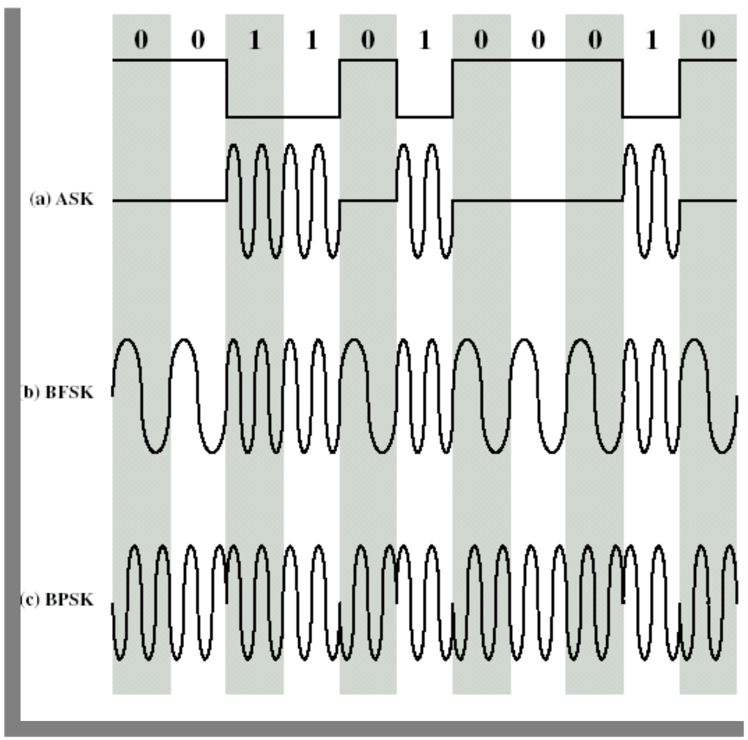
  频带利用率是指单位频带内的传输速率
- **误码率**

  在传输过程中发生误码的码元个数与传输的总码元数之比

调制和解调



Carrier signal



Modulated signal

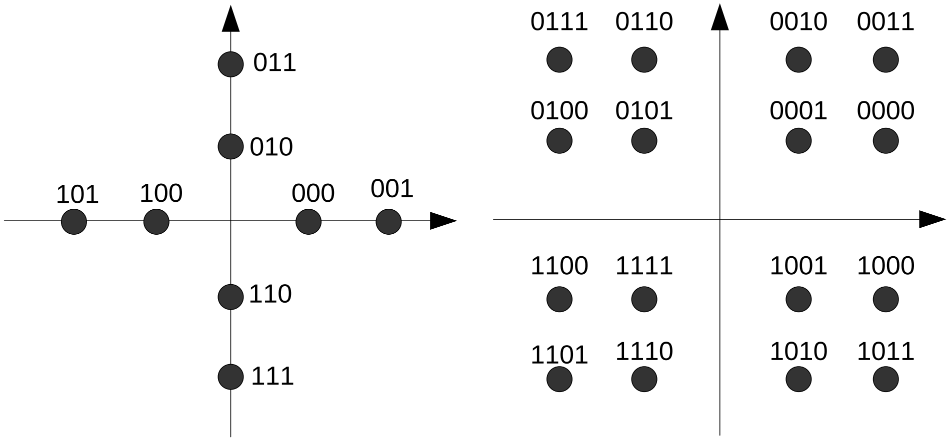
$$x(t) = A(t)\cos\left[\omega_c t + \phi(t)\right]$$

Amplitude modulation (AM)...,
Amplitude Shift Keying (ASK)...

Frequency modulation (FM),
Frequency/Phase Shift Keying (FSK,PSK)...

数字调制技术

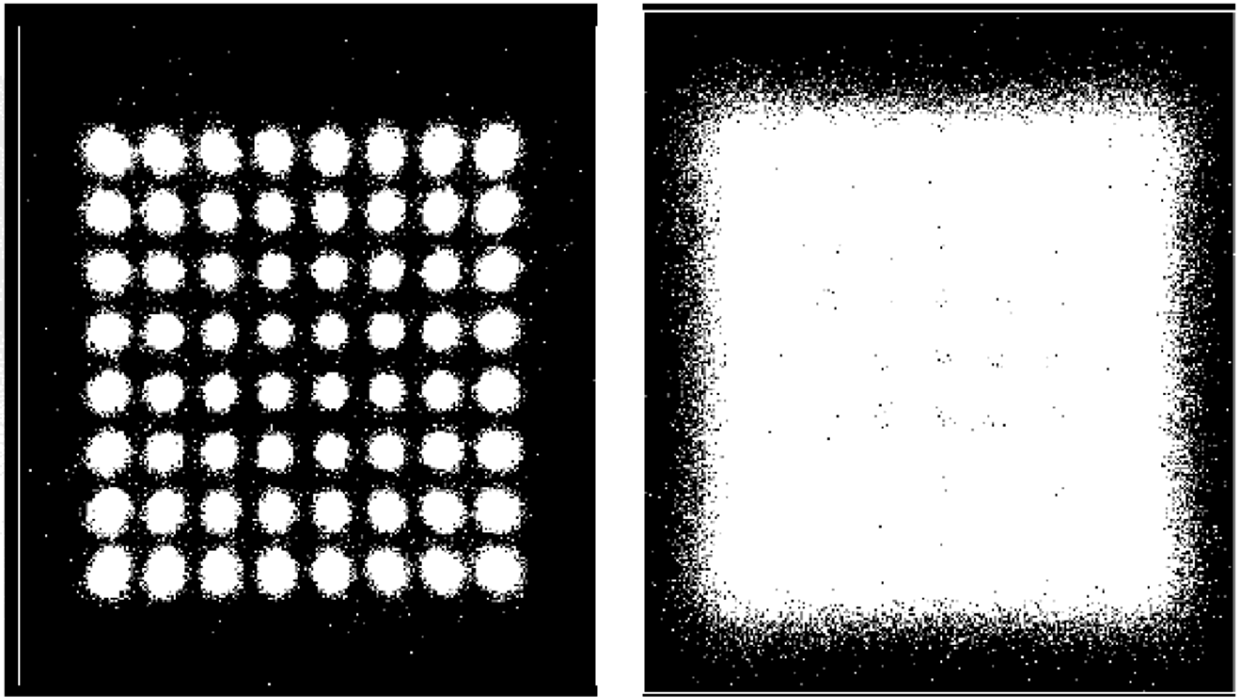
- QAM (Quadrature Amplitude Modulation)
  就是调相和调幅组合



8-QAM 16-QAM

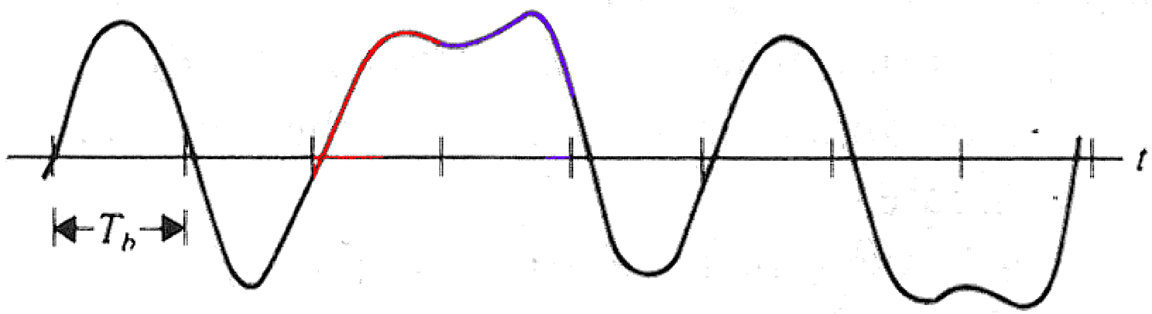
数字调制实例

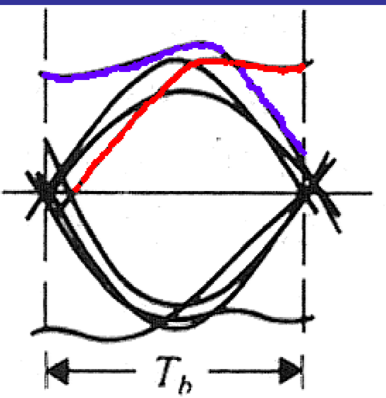
- 不同干扰下的接收星座图

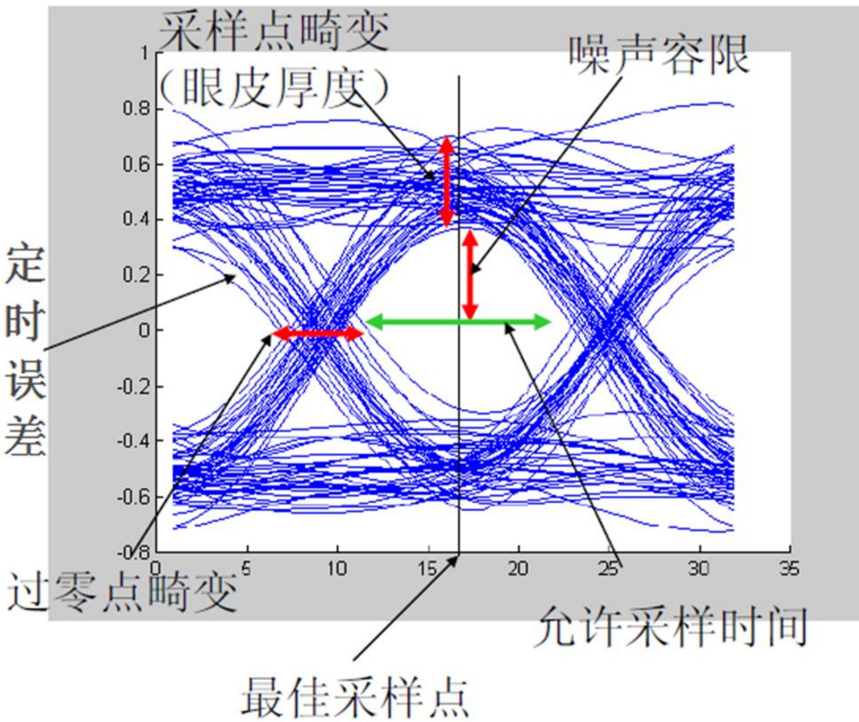


（a）噪声较小　　　（b）噪声较大

传输质量



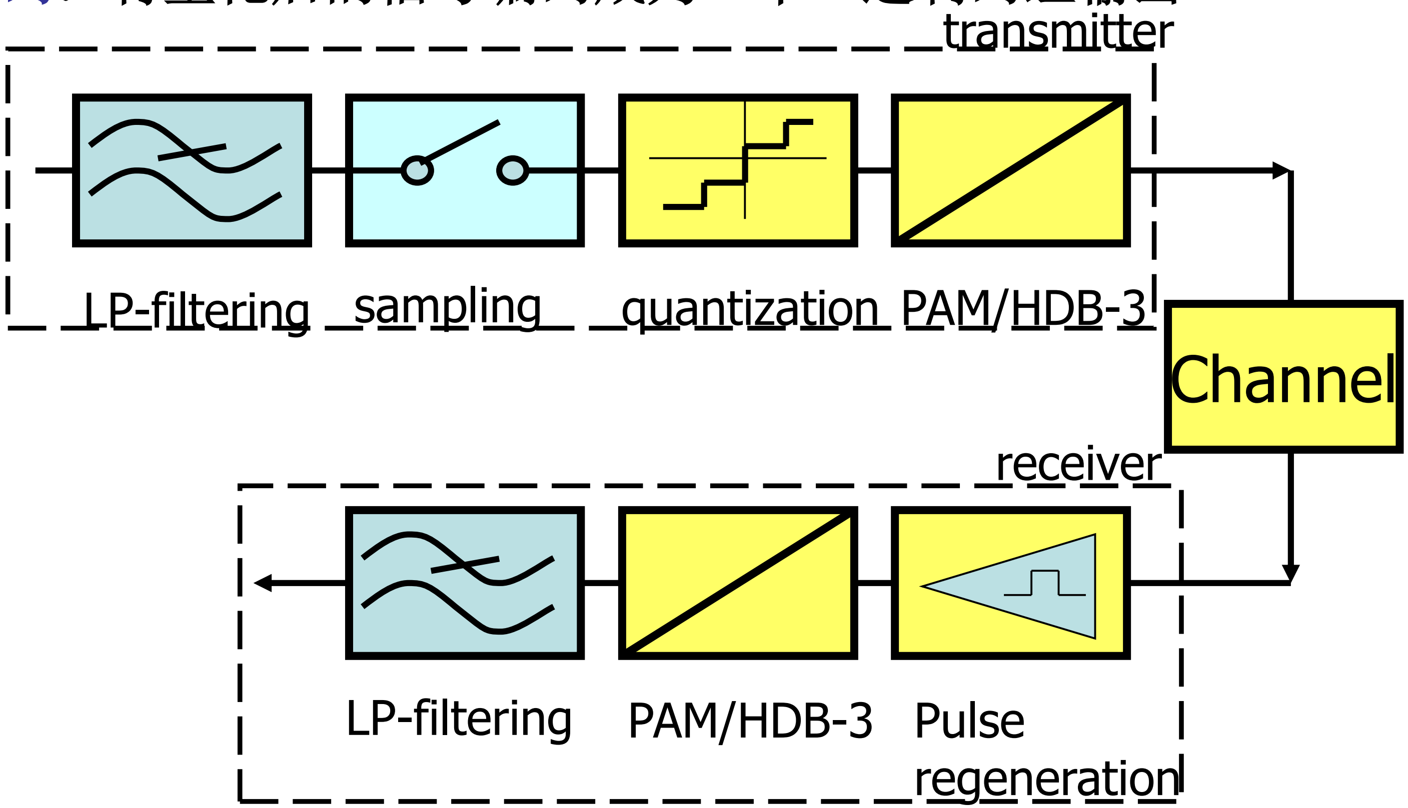




- 最佳采样点
- 采样点畸变
- 允许采样时间
- 噪声容限
- 过零点畸变
- 定时误差

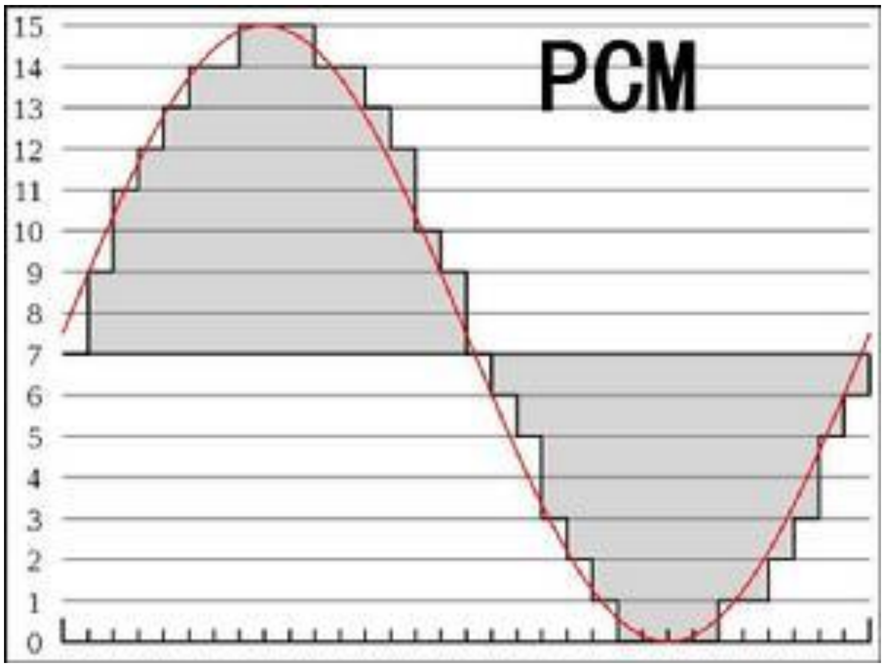
脉冲编码调制 (PCM)

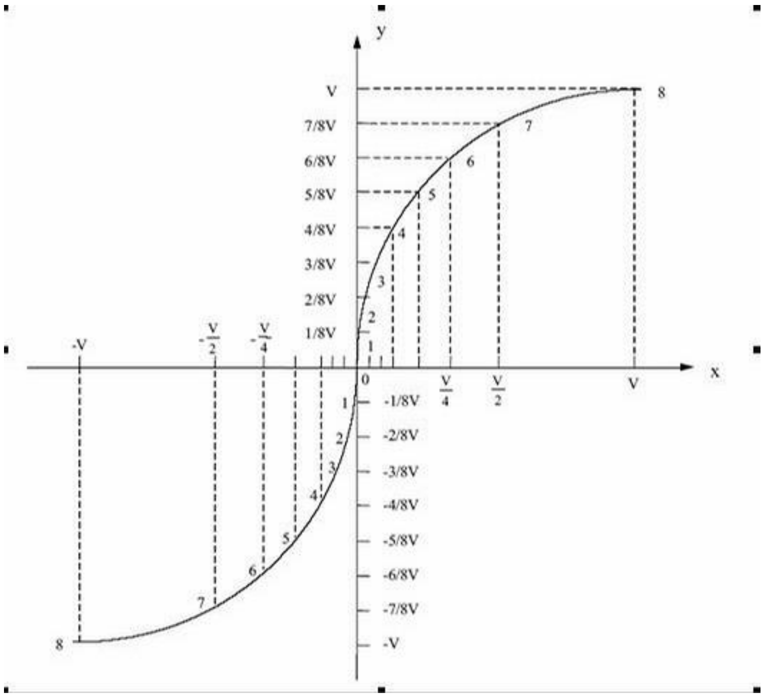
- **抽样**：将连续时间模拟信号变为离散时间、连续幅度的信号
- **量化**：将抽样信号变为离散时间、离散幅度的数字信号
- **编码**：将量化后的信号编码成为一个二进制码组输出



脉冲编码调制 (PCM)

- 均匀量化和非均匀量化

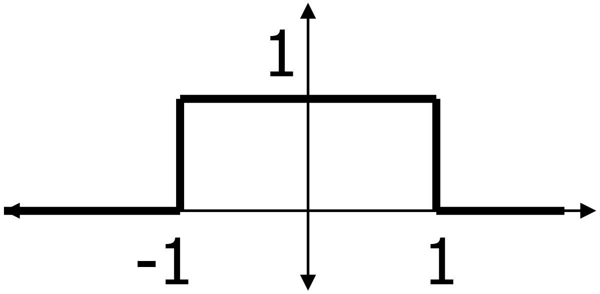




通信理论

傅里叶变换

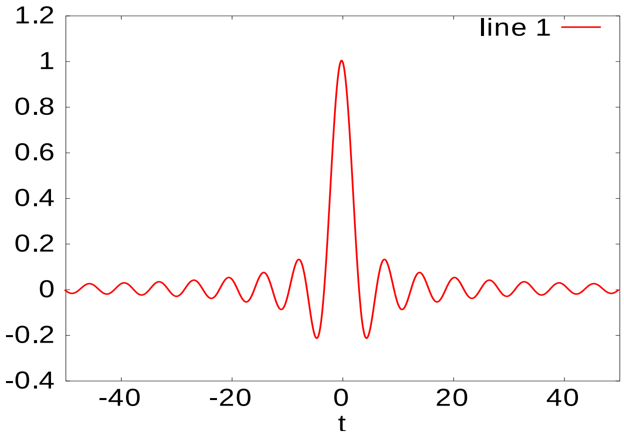
$$x(t) = \begin{cases} 1 & \text{for } t \in [-1, 1] \\ 0 & \text{otherwise} \end{cases}$$



$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt$$

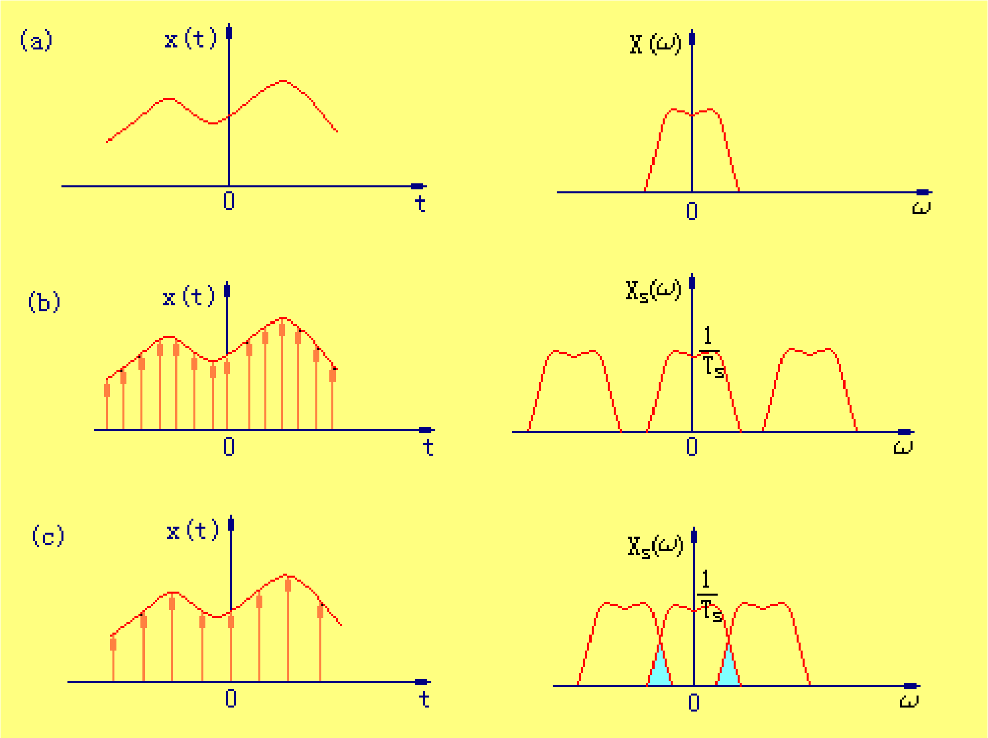
$$= \frac{1}{-i\omega} e^{-i\omega t} \Big|_{t=-1}^{1}$$

$$= \frac{1}{-i\omega} \left(-2i \sin(\omega) \right)$$



奈奎斯特采样定理

- 在进行模拟/数字信号的转换过程中，当采样频率大于信号中最高频率2倍时，采样之后的数字信号完整地保留了原始信号中的信息



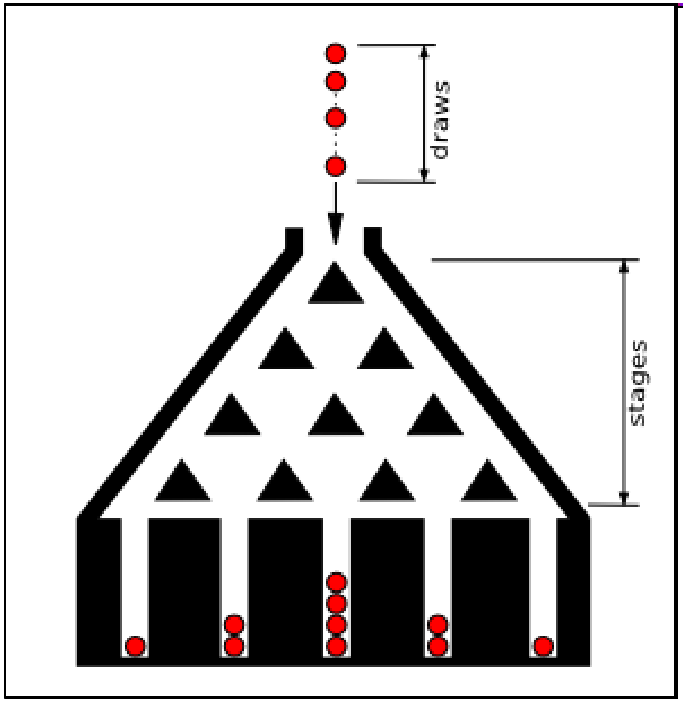
大数定理和概率的定义

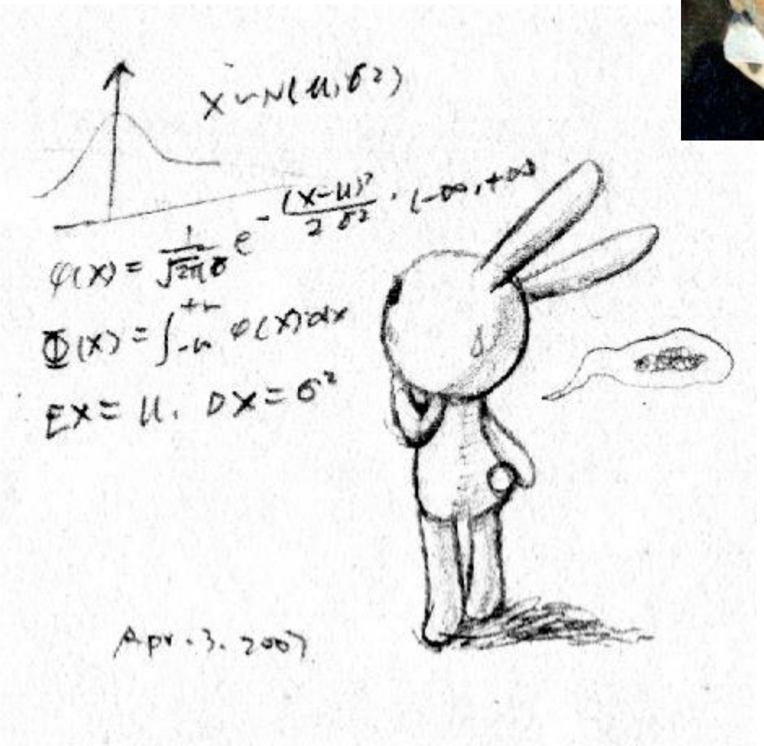
The more repetitions, the better the approximation $p \simeq f/n$.

This is sometimes referred to as the Law of Large Numbers, which states that if an experiment is repeated a large number of times, the relative frequency of the outcome will tend to be close to the probability of the outcome.

Shortly, we will consider another approach to determining the probability of an event based on logical reasoning.

中心极限定理-正态分布


draws
stages


$X \sim N(\mu, \sigma^2)$
$\varphi(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, (-\infty, +\infty)$
$\Phi(x) = \int_{-\infty}^{x} \varphi(x)dx$
$EX = \mu, DX = \sigma^2$
Apr. 3. 2007

贝叶斯准则

Rev. T. Bayes

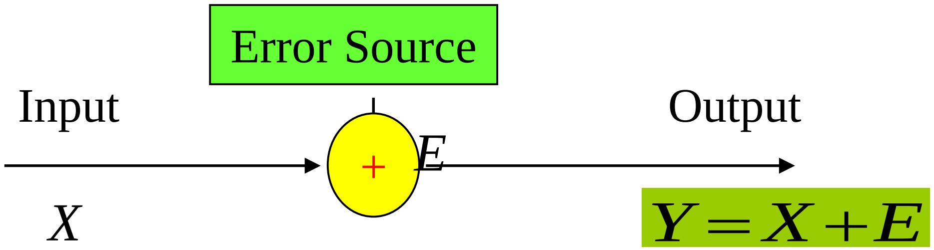
描述两个条件概率之间的关系

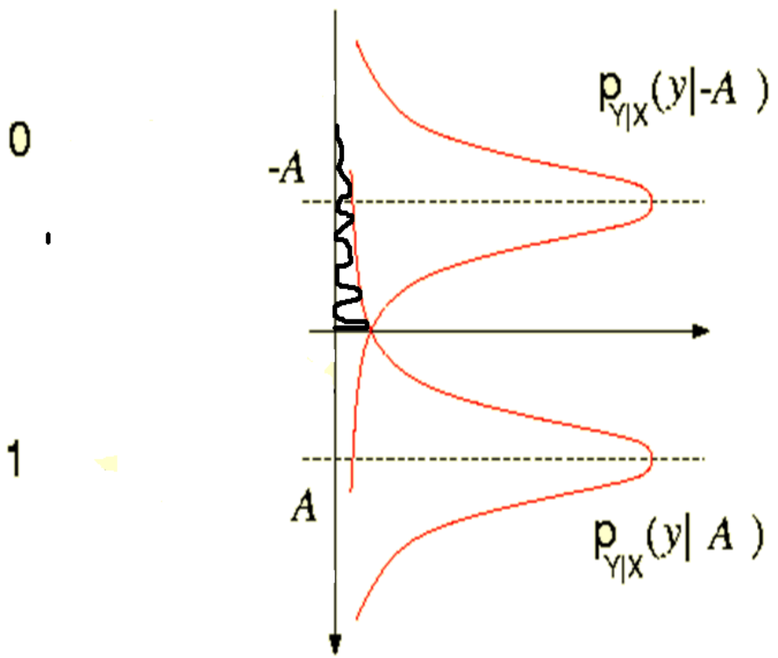
$P(A\cap B)=P(A)\cdot P(B|A)=P(B)\cdot P(A|B)$

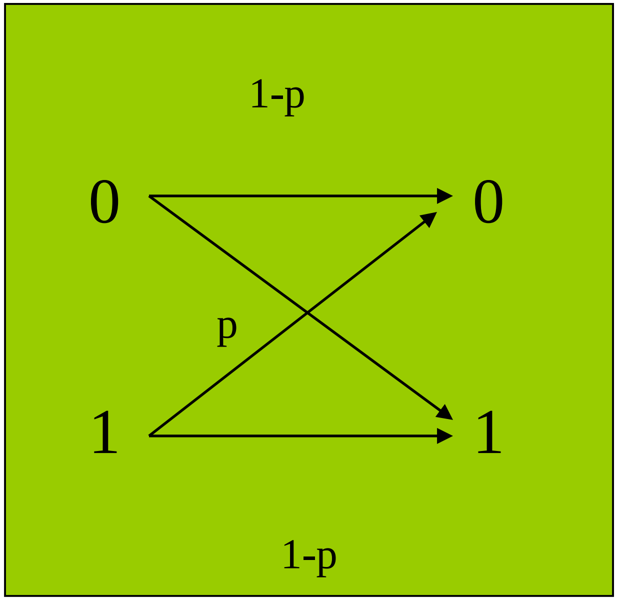
一座别墅在过去的 20 年里一共发生过 7 次被盗，别墅的主人有一条狗，狗平均每周晚上叫 3 次，在盗贼入侵时狗叫的概率被估计为 0.8，问题是：在狗叫的时候发生入侵的概率是多少？

我们假设 A 事件为狗在晚上叫，B 为盗贼入侵，则 $P(A) = 3/7$，$P(B)=7/(20\cdot 365)=2/7300$，$P(A|B) = 0.8$，按照公式很容易得出结果(5.11E-4!)

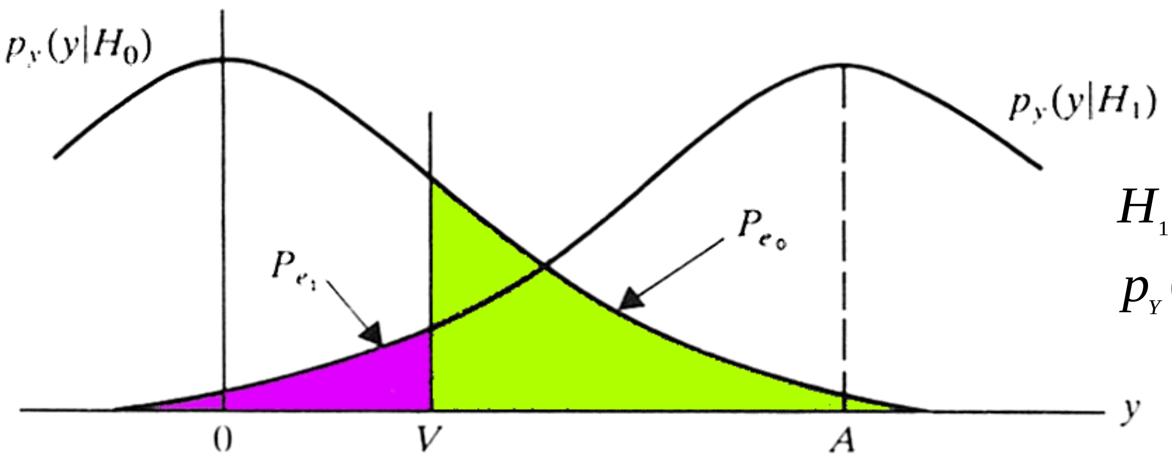
检测理论


Error Source
Input
Output
E
X
Y = X + E


0
1
-A
A
p_Y|X(y|-A)
p_Y|X(y| A)


1-p
0
0
p
1
1
1-p

确定判决门限



$$H_0: a_k = 0, Y = n$$
$$p_Y(y \mid H_0) = p_N(y)$$
$$H_1: a_k = 1, Y = A + n$$
$$p_Y(y \mid H_1) = p_N(y - A)$$

The comparator implements decision rule:

Choose Ho *(a_k=0)* if *Y<V*
Choose H1 *(a_k=1)* if *Y>V*

$$p_{e1} \equiv P(Y < V \mid H_1) = \int_{-\infty}^{V} p_Y(y \mid H_1)dy$$

$$p_{eo} \equiv P(Y > V \mid H_0) = \int_{V}^{\infty} p_Y(y \mid H_0)dy$$

Transmitted ‘0’ but detected as ‘1’

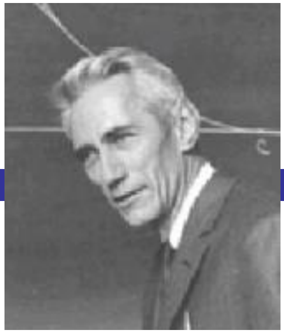
Average error error probability: $P_e = P_0 P_{e0} + P_1 P_{e1}$

$P_0 = P_1 = 1/2 \Rightarrow P_e = ½(P_{e0} + P_{e1})$

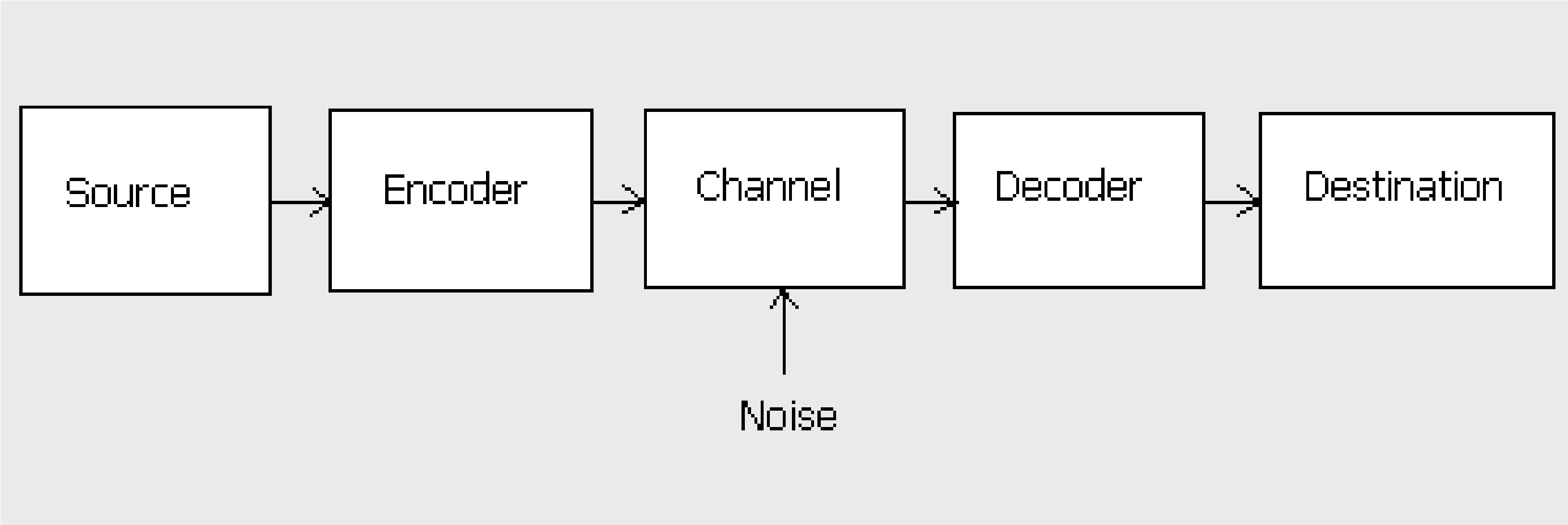
Assume Gaussian noise:

$$p_N(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2}\right)$$

信息论



- Communication theory deals with systems for transmitting information from one point to another.



- Information theory was born with the discovery of the fundamental laws of data compression and transmission.
- Shannon uses the definition of entropy $$H = -\sum_{i=1}^{n} p_i \log p_i$$

信道容量

Definition:

The rate R of a code is the ratio k/n, where

k is the number of information bits transmitted in n channel uses

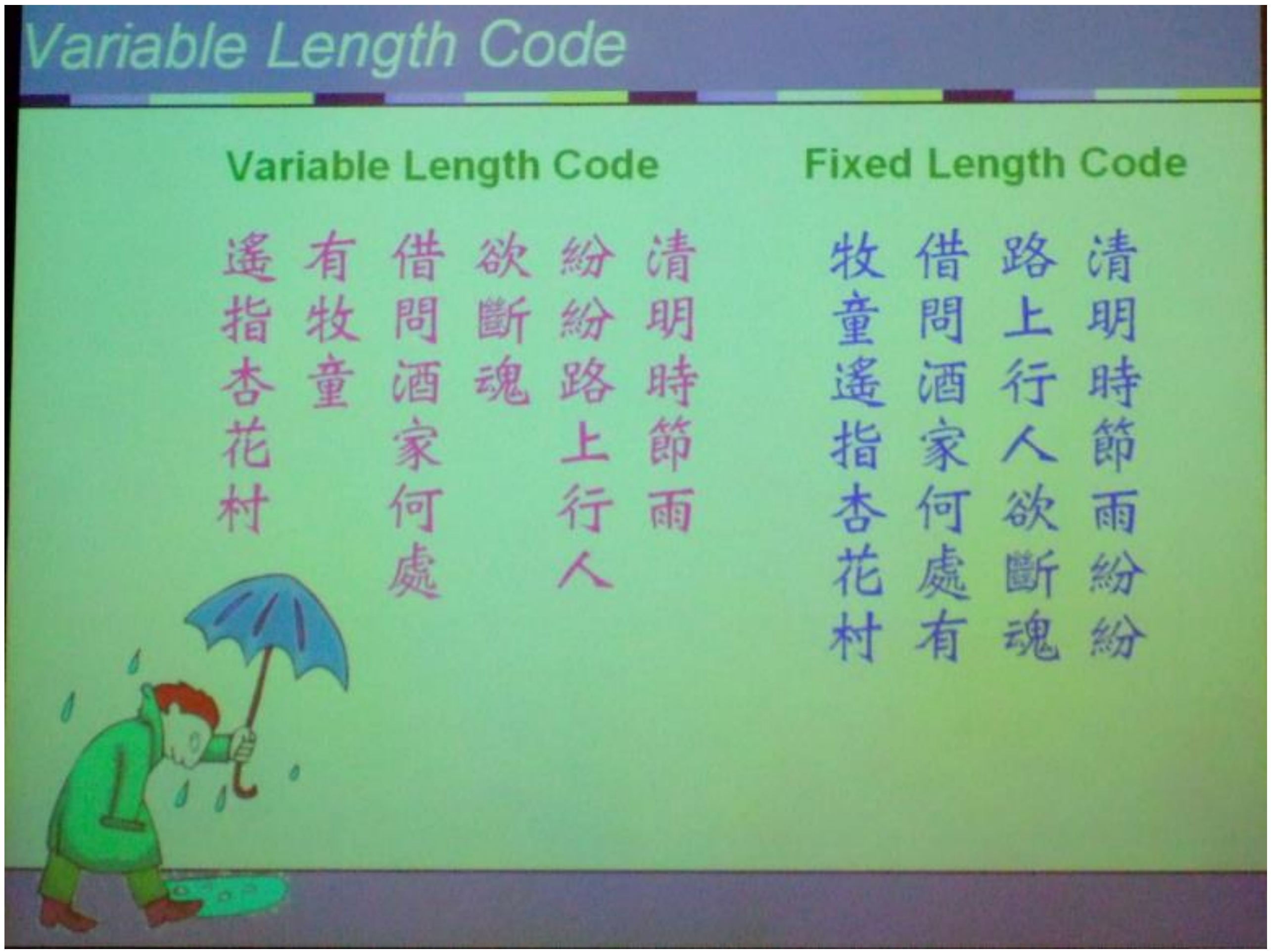
Shannon showed that: :

for $R \le C$

encoding methods exist

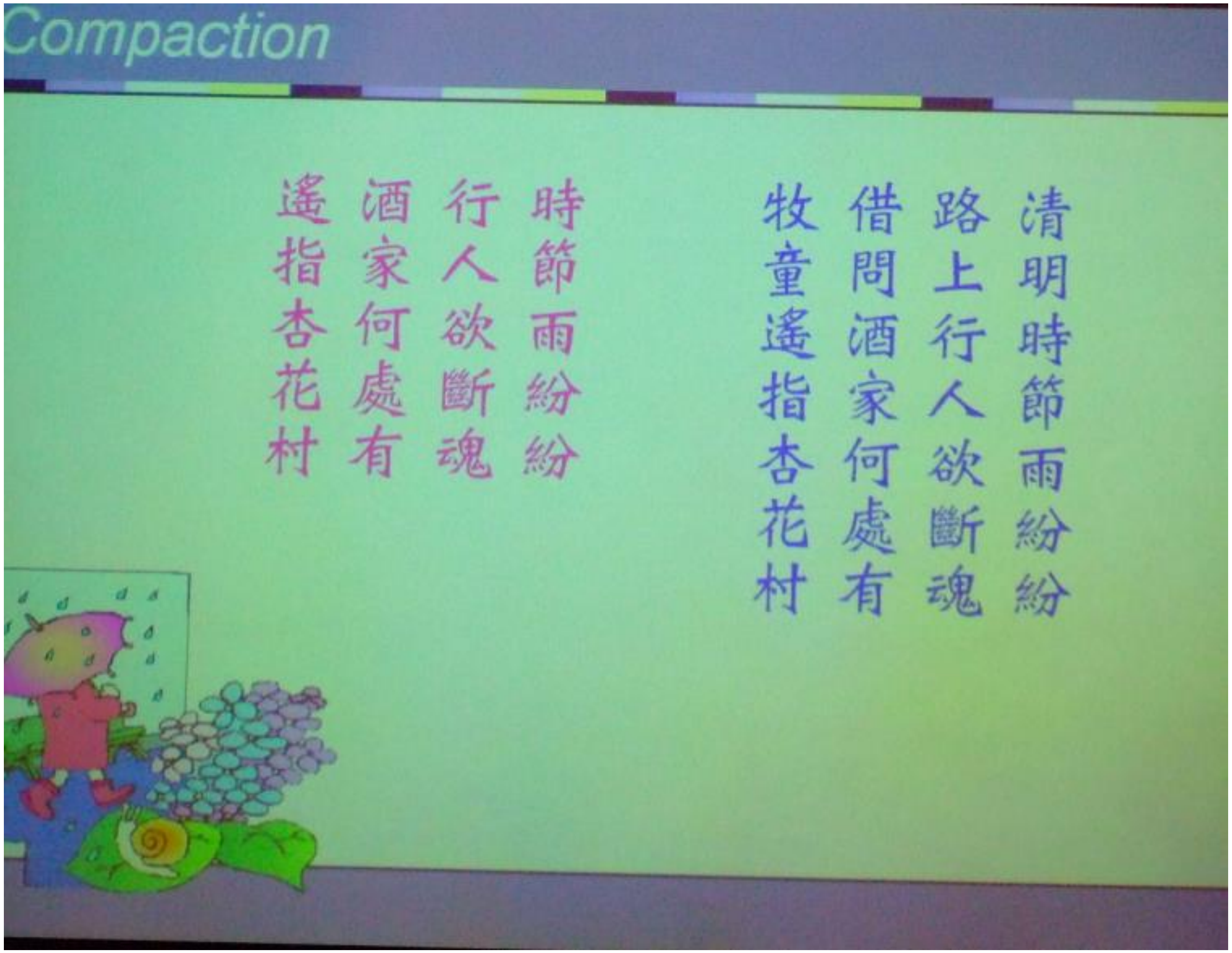
with decoding error probability ⇨ 0

变长编码



Variable Length Code
Variable Length Code
清明時節雨
紛紛路上行人
欲斷魂
借問酒家何處
有牧童
遙指杏花村
Fixed Length Code
清明時節雨紛紛
路上行人欲斷魂
借問酒家何處有
牧童遙指杏花村

压缩编码


Compaction
時節雨紛紛
行人欲斷魂
酒家何處有
遙指杏花村
清明時節雨紛紛
路上行人欲斷魂
借問酒家何處有
牧童遙指杏花村

纠错编码


Representation of Information : Reliability
Repetition 疊字
尋尋覓覓，冷冷清清，淒淒慘慘戚戚。
乍暖還寒時候，最難將息。
三杯兩盞淡酒，怎敵他，晚來風急。
雁過也，正傷心，卻是舊時相識。
滿地黃花堆積，憔悴損，如今有誰堪摘。
守著窗兒，獨自怎生得黑，
梧桐更兼細雨，到黃昏，點點滴滴。
這次第，怎一個愁字了得。
〈聲聲慢 李清照〉

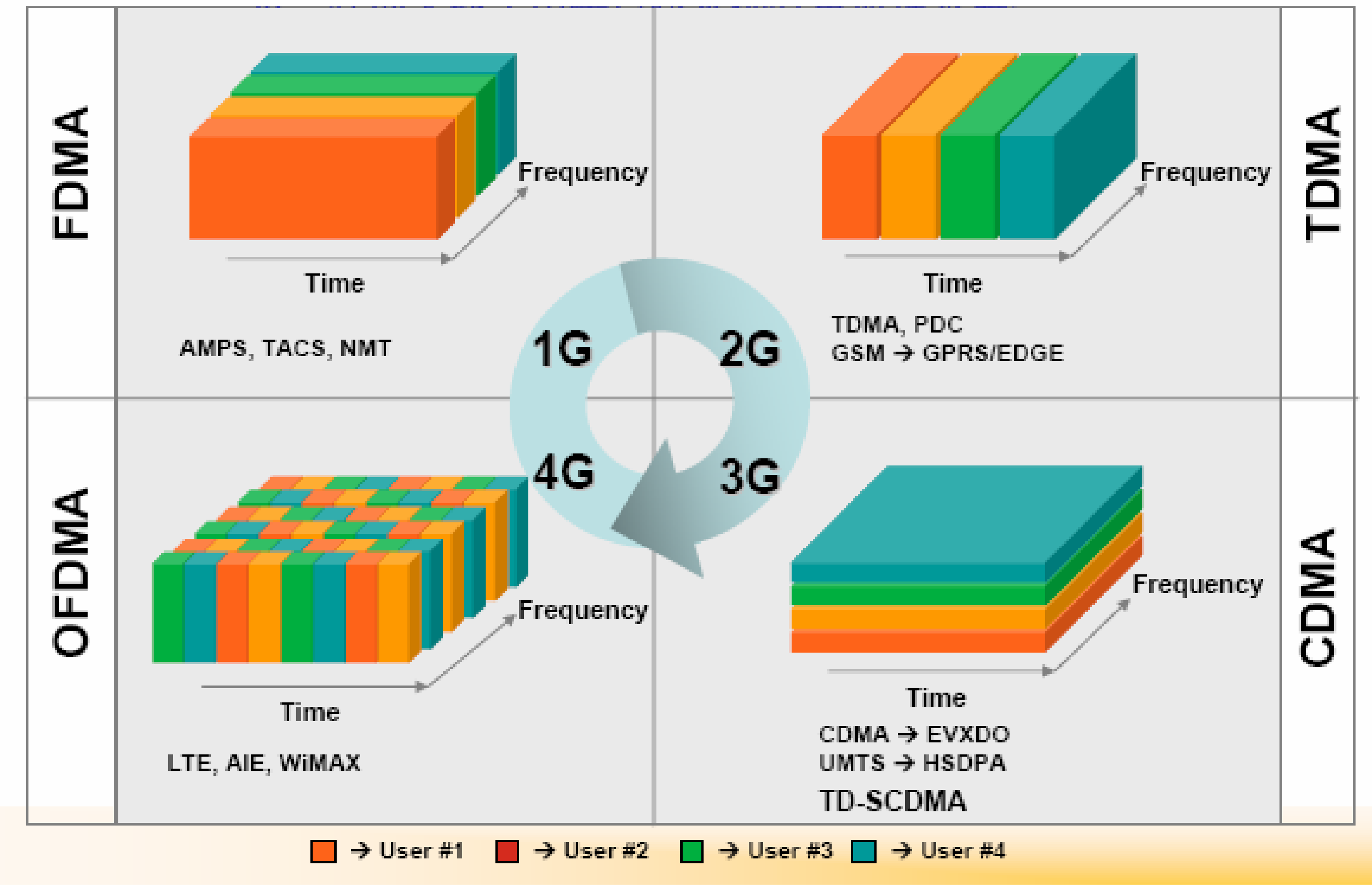
加密编码

加密编码


Representation of Information : Security
四 猜成語一句
欲罷不能
四 猜唐詩一句
山在虛無縹緲間

多址方式

频分/时分/码分复用

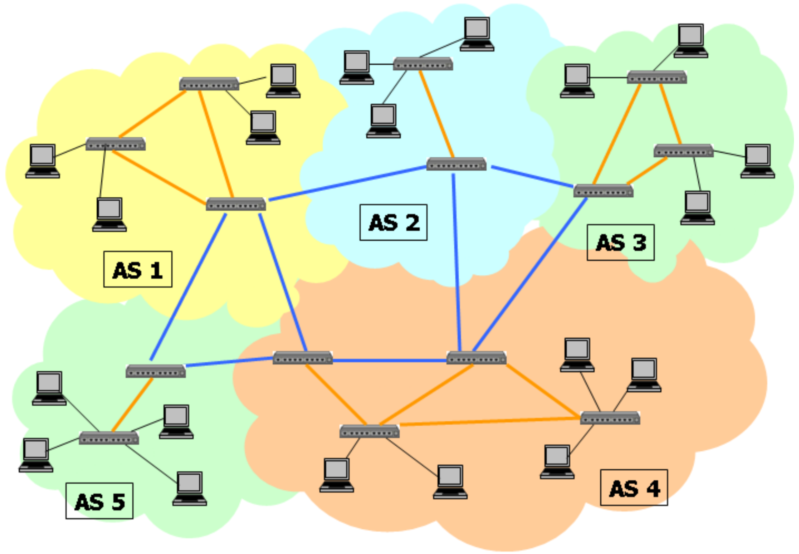


FDMA
Frequency
Time
AMPS, TACS, NMT
TDMA
Frequency
Time
TDMA, PDC
GSM → GPRS/EDGE
1G
2G
3G
4G
OFDMA
Frequency
Time
LTE, AIE, WiMAX
CDMA
Frequency
Time
CDMA → EVXDO
UMTS → HSDPA
TD-SCDMA
→ User #1
→ User #2
→ User #3
→ User #4

网络协议

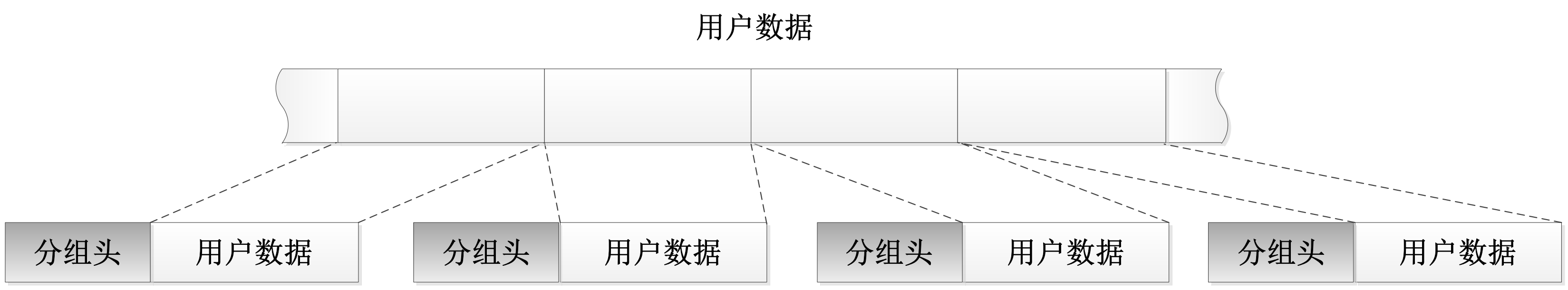
交换

- 电路交换
  - 独占一条物理线路
- 报文交换
  - 存储和转发
- 分组交换
  - 分组交换在线路上采用动态复用技术传送按一定长度分割为许多小段的数据



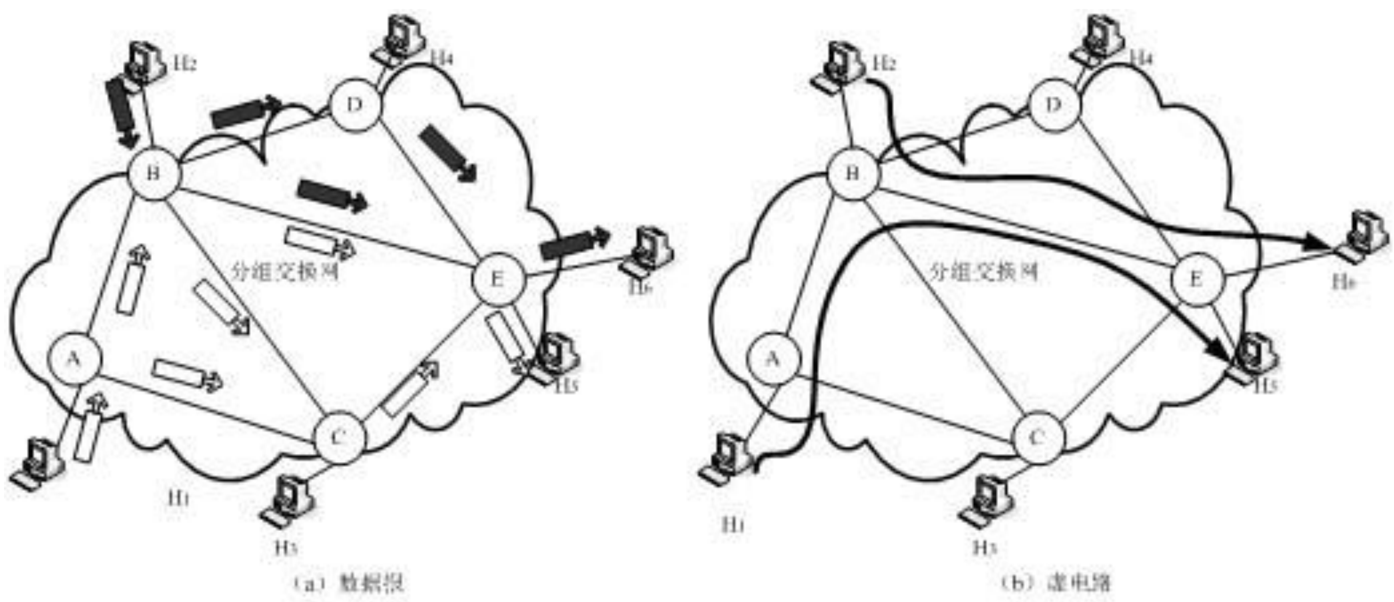
Packet Switching: A closer look

- **Objective**
  - **High utilization of communication links**
- **Bandwidth efficiency: Statistical multiplexing**
  - **Buffers smoothen peak traffic load**



Packet Switching Alternatives

- ➢Virtual-circuit transport （虚电路）
- ➢Datagram transport （数据报）

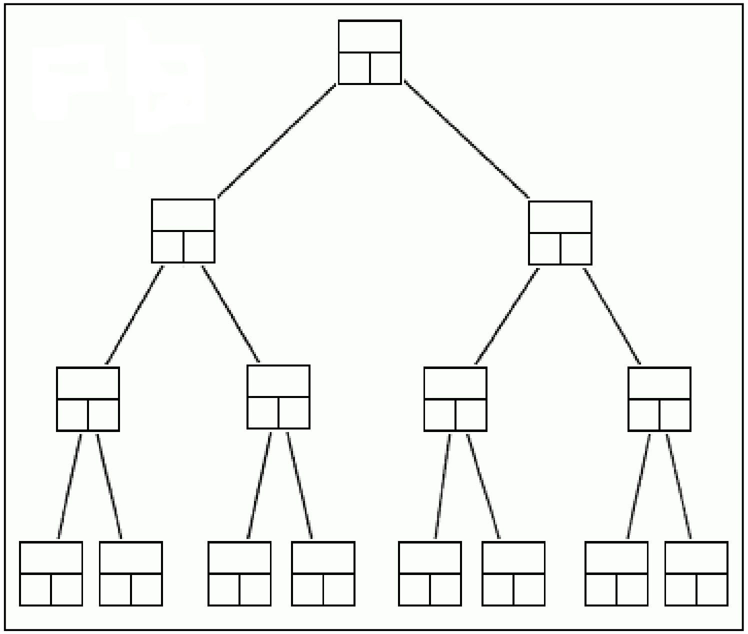


Data Structures

A data structure is a scheme for organizing data in the memory of a computer.

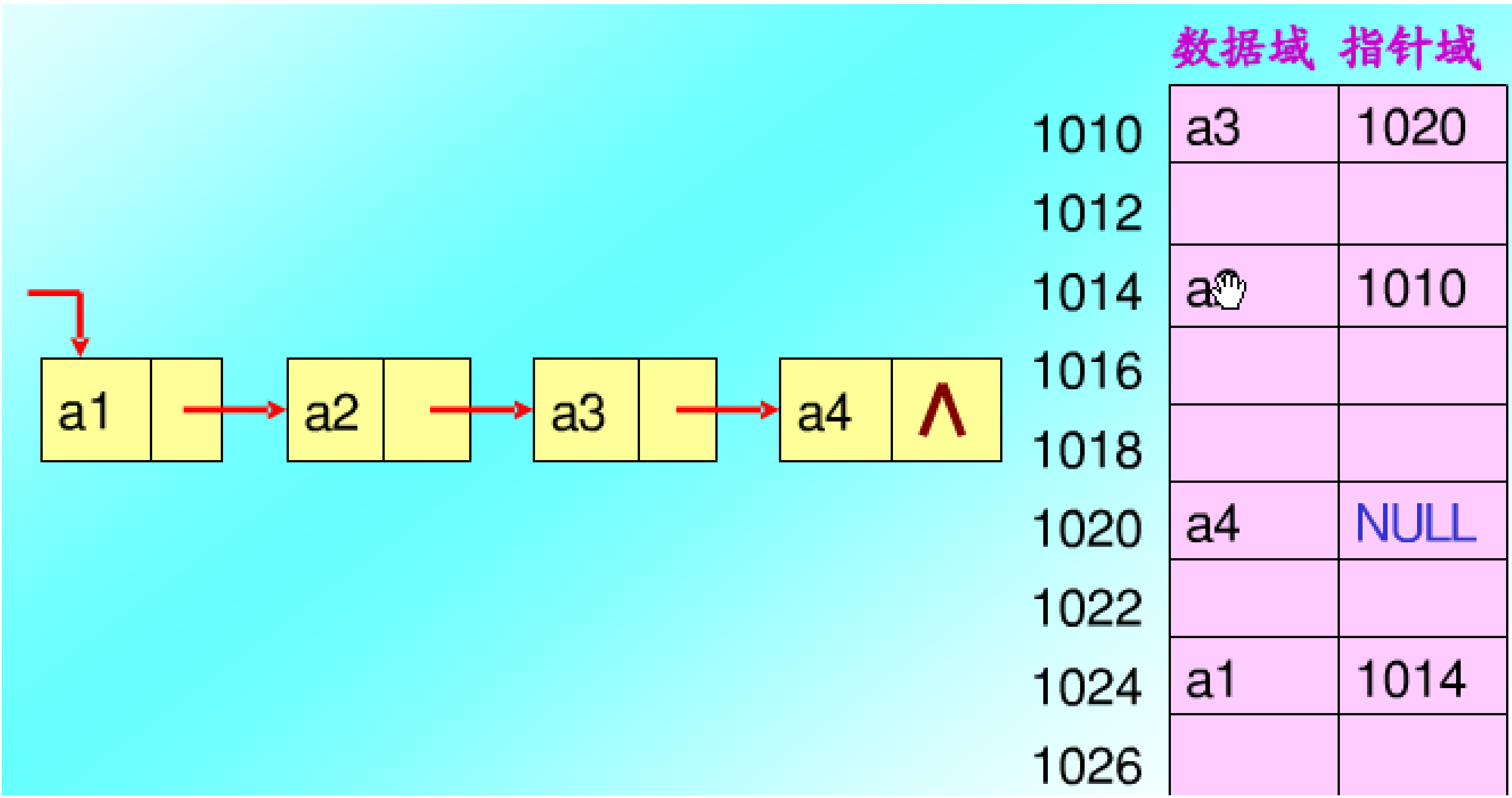
Some of the more commonly used data structures include lists, arrays, stacks, queues, heaps, trees, and graphs.

Computer programmers decide which data structures to use based on the nature of the data and the processes that need to be performed on that data.



Binary Tree

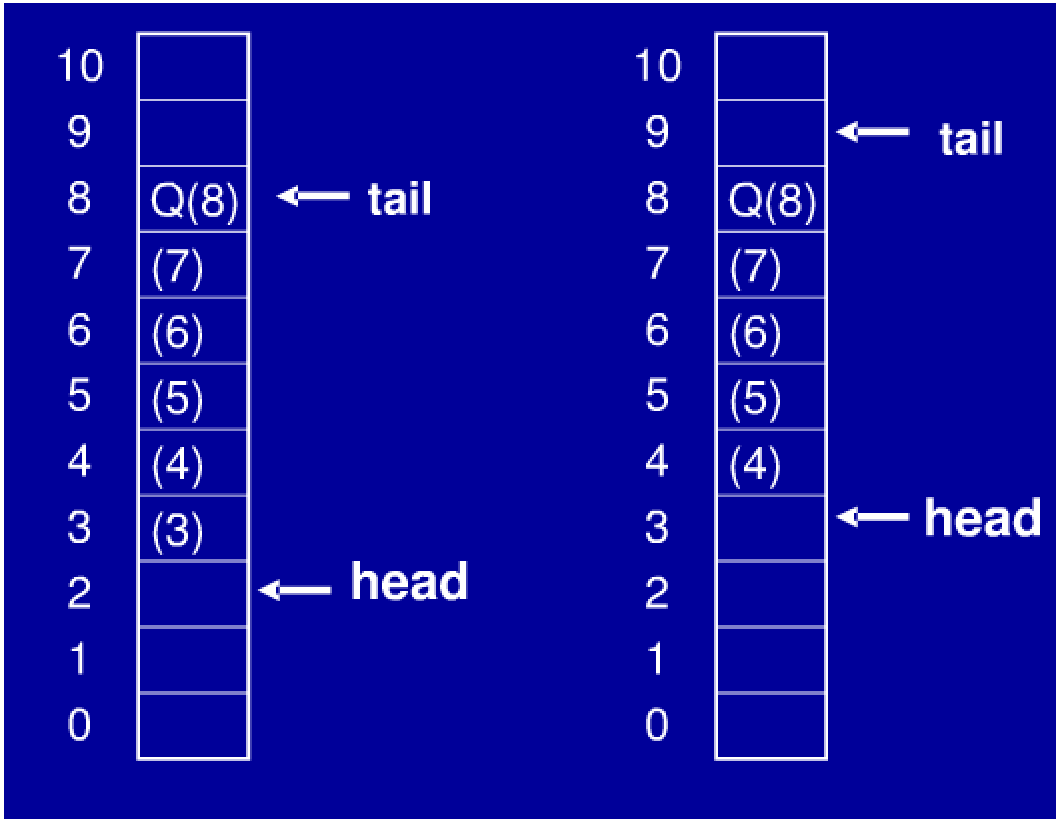
链表



a1
a2
a3
a4
∧
数据域
指针域
1010
1012
1014
1016
1018
1020
1022
1024
1026
a3
1020
1010
a4
NULL
a1
1014

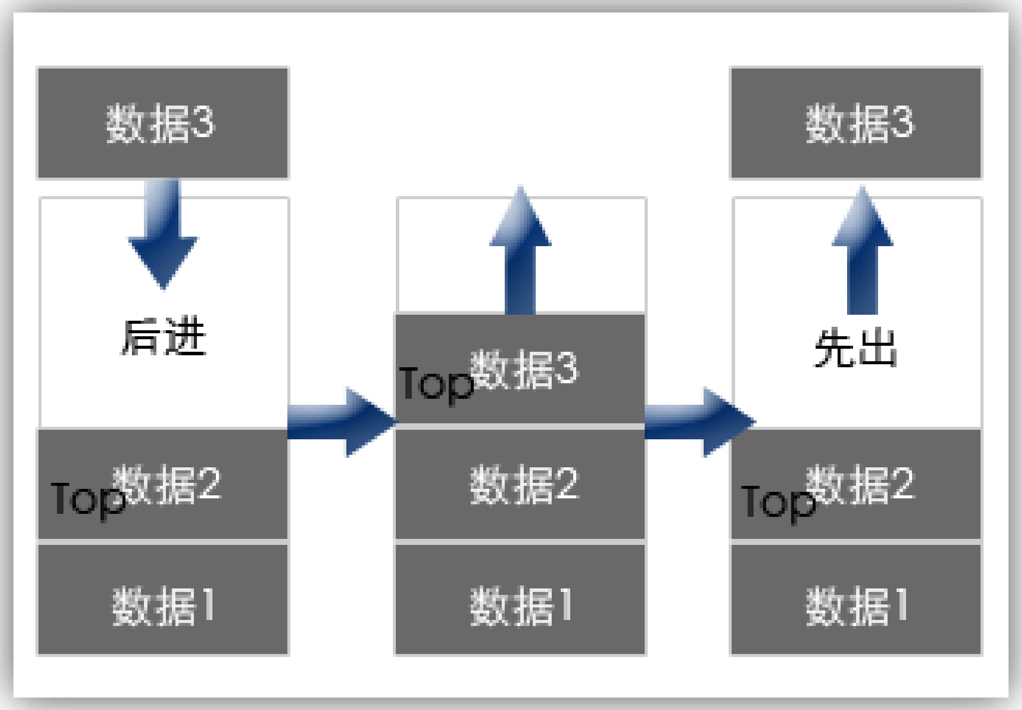
队列(FIFO)

删除只能在表头,插入只能在表尾



堆栈(LIFO)

用后进先出的方式来描述数据的存取方式



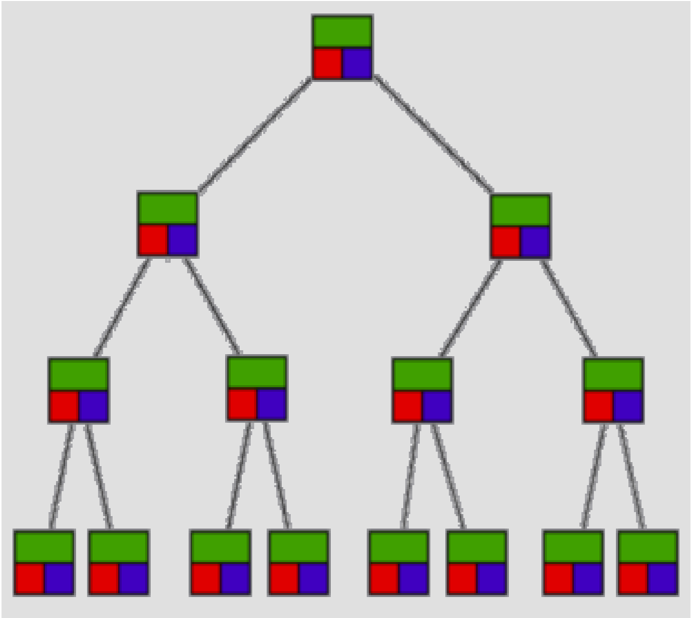
Choosing Data Structures

For some applications, a queue is the best data structure to use.

For others, a binary tree is better.

Programmers choose from among many data structures based on how the data will be used by the program.

A *queue* is an example of commonly used simple data structure. A queue has beginning and end, called the *front* and *back* of the queue. - printing



A *binary tree* is another commonly used data structure. It is organized like an upside down tree.
Each spot on the tree, called a *node*, holds an item of data along with a left pointer and a right pointer. - sorting

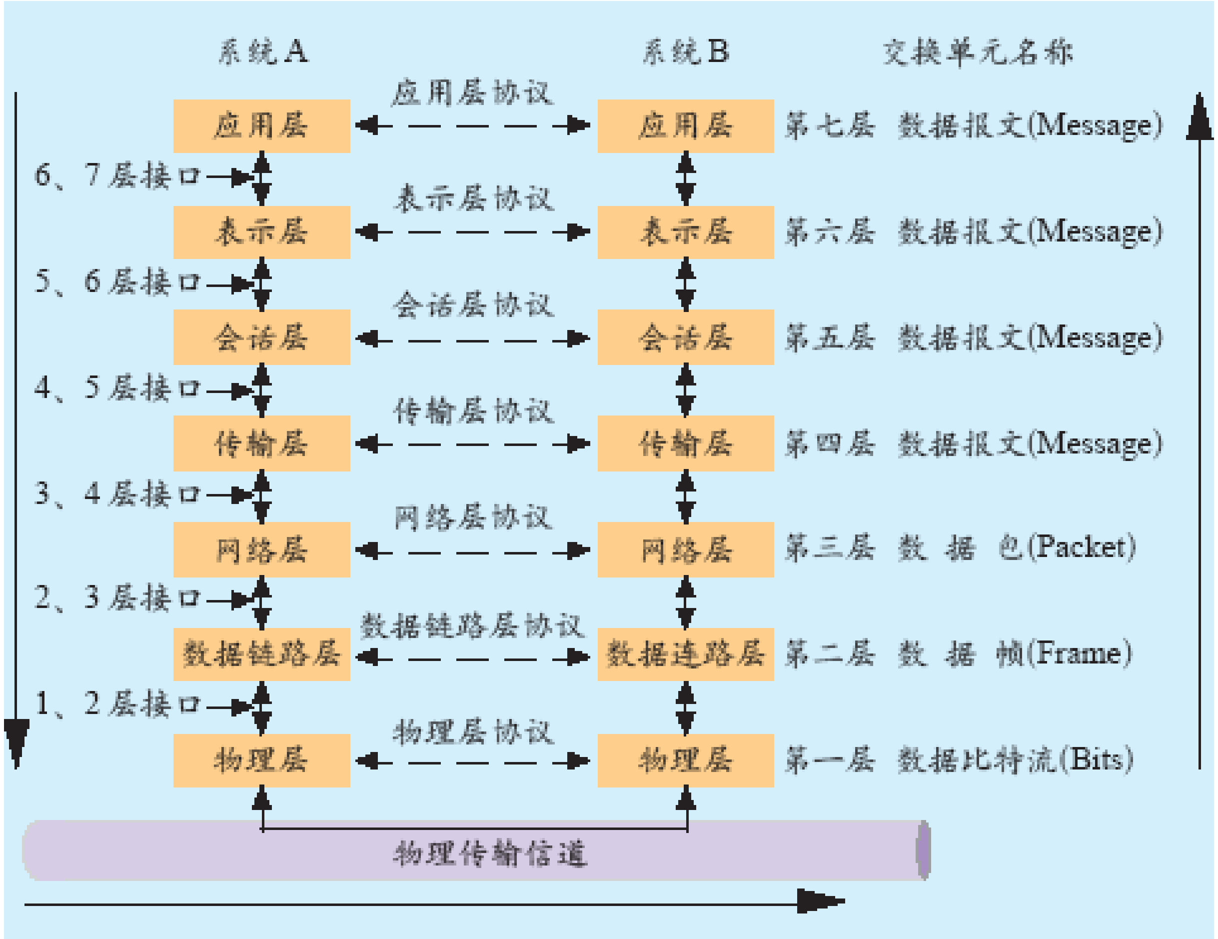
网络分层结构

- Information/ Complexity hiding
- One Layer changes do not affect others

ideas		ideas
languages		languages
medium		medium

网络分层结构

- 服务
- 接口
- 协议



网络实例

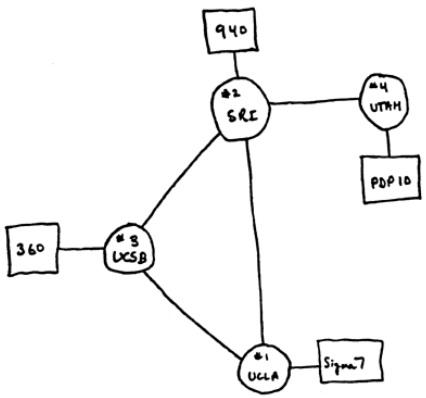
INTERNET

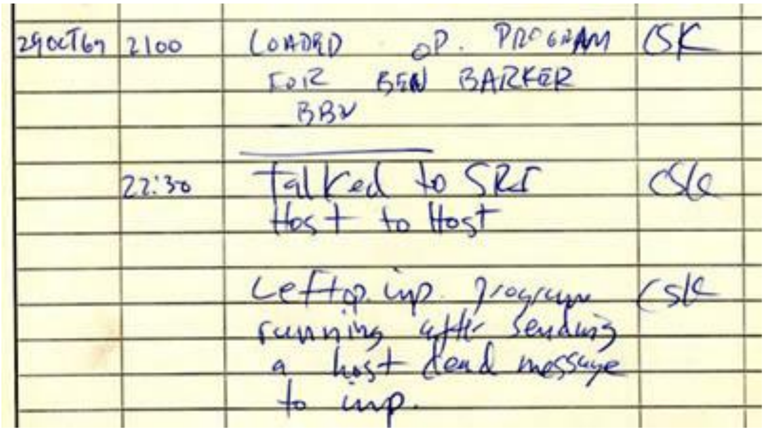
INTERNET历史

- **Big question**
  - **How to build a reliable networking using unreliable components?**
- **Theory**
  - **Kleinrock L.Information flow in large communication nets: [Ph D Thesis Proposal]. Massachusetts Institute of Technology, Cambridge, May 1961**
  - **Paul Baran seminal work first appeared in a series of RAND studies published between 1960 and 1962 and then finally in the tome "On Distributed Communications," published in 1964.**

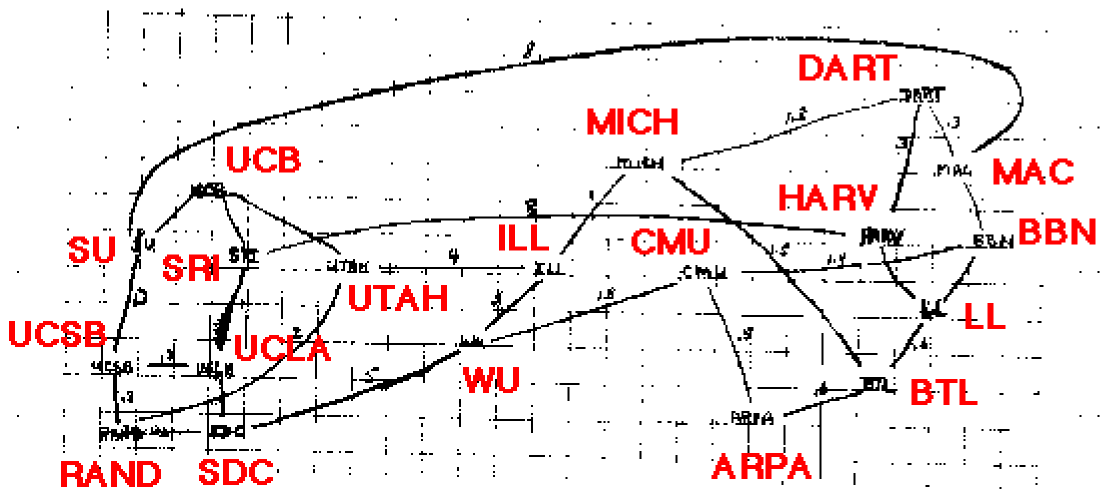
ARPANET

Larry Robert





Interface Message Processor (IMP)



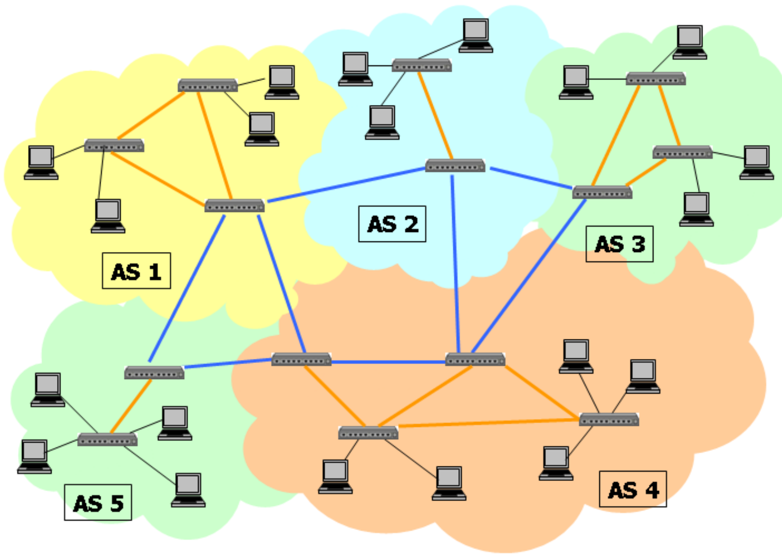
TCP/IP协议

- Vinton G. Cerf and Robert Kahn

  – Concept in 1972
  – Paper in 1974
  – Implementation in 1978
  – Industry standard in 1983

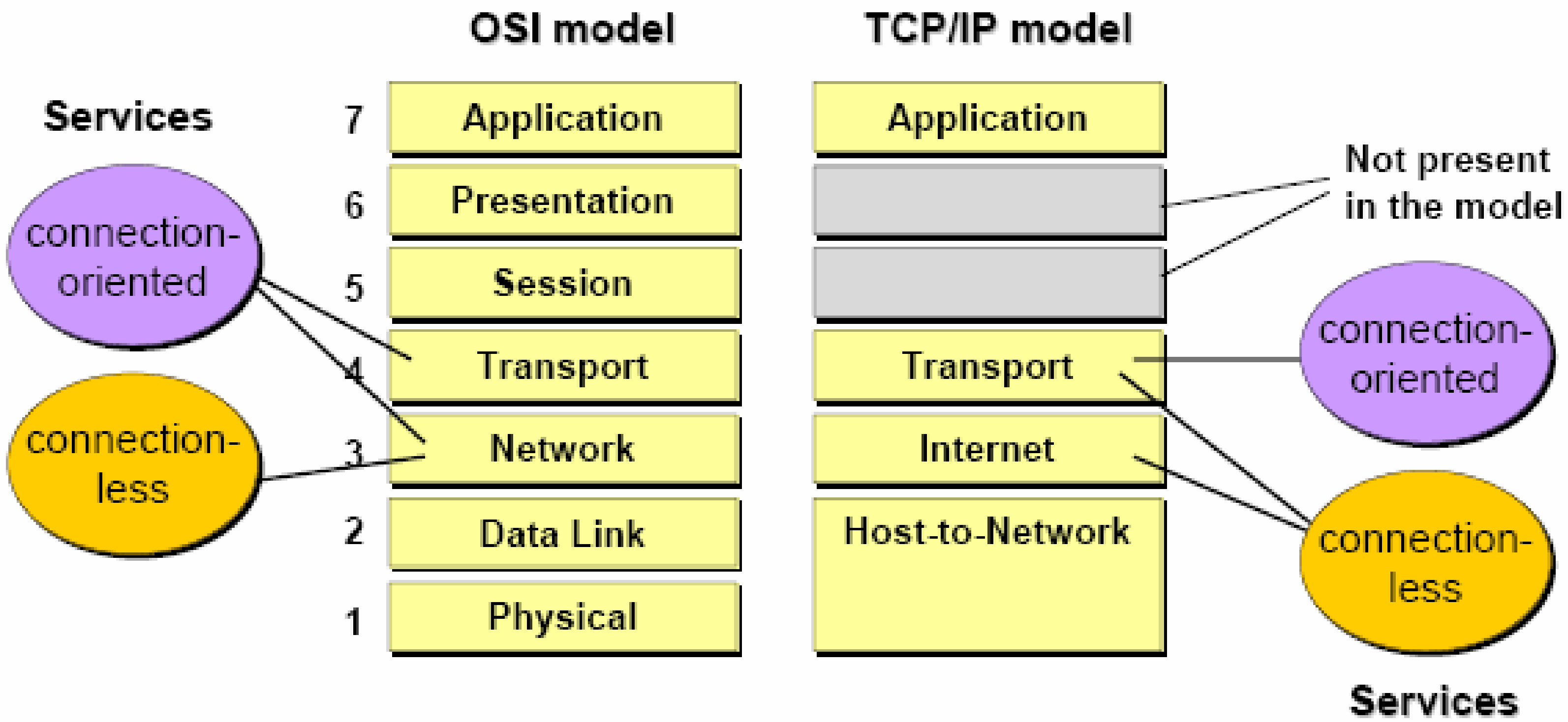
Internet的系统结构

- 无连接 (connectionless)
- 端对端(end-to-end)
- 尽力而为 (best effort)

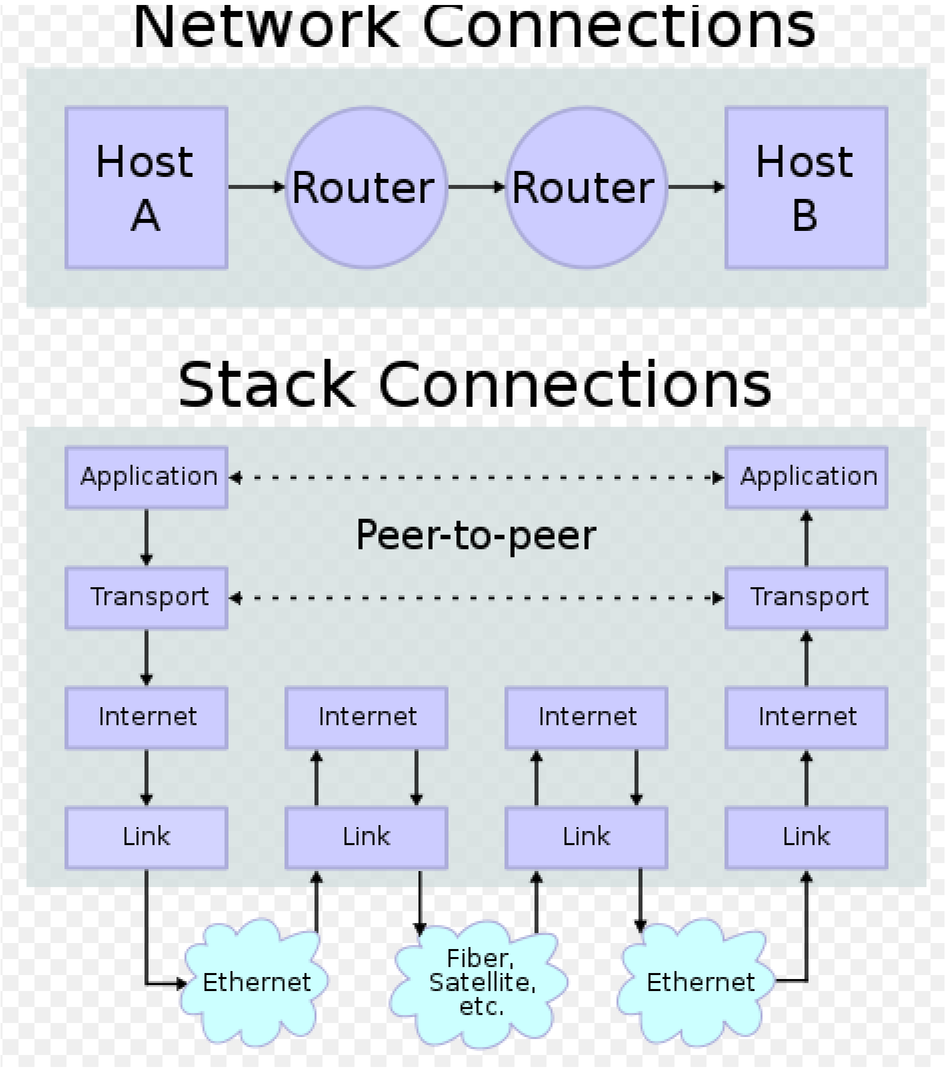


The Internet Architecture

Comparison of OSI and TCP/IP model:

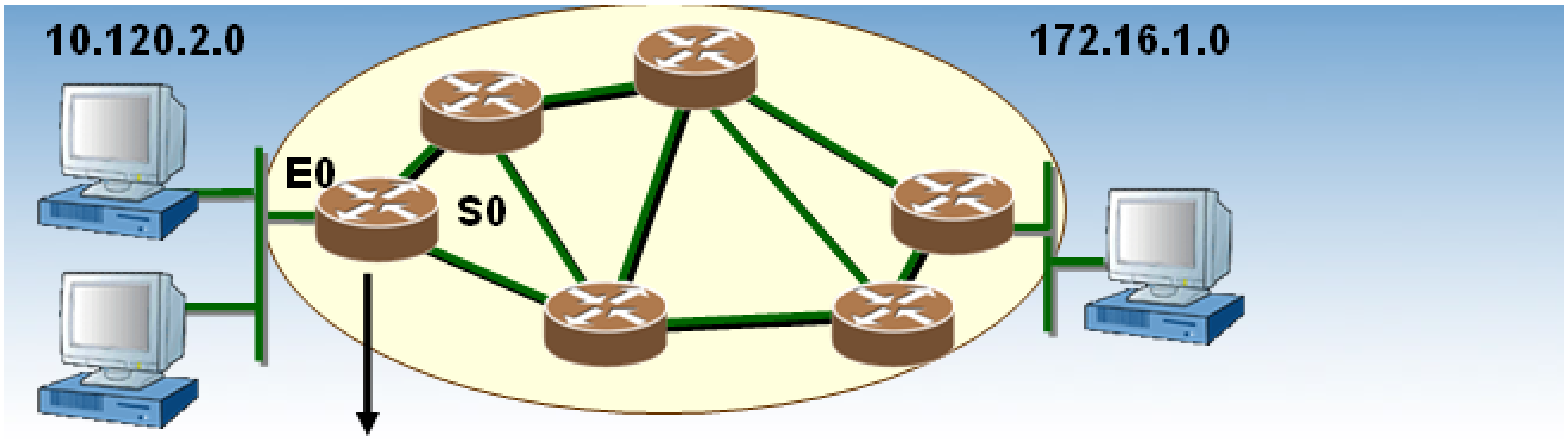


IP连接和分层结构



Network Connections
Host A
Router
Router
Host B
Stack Connections
Application
Application
Peer-to-peer
Transport
Transport
Internet
Internet
Internet
Internet
Link
Link
Link
Link
Ethernet
Fiber, Satellite, etc.
Ethernet

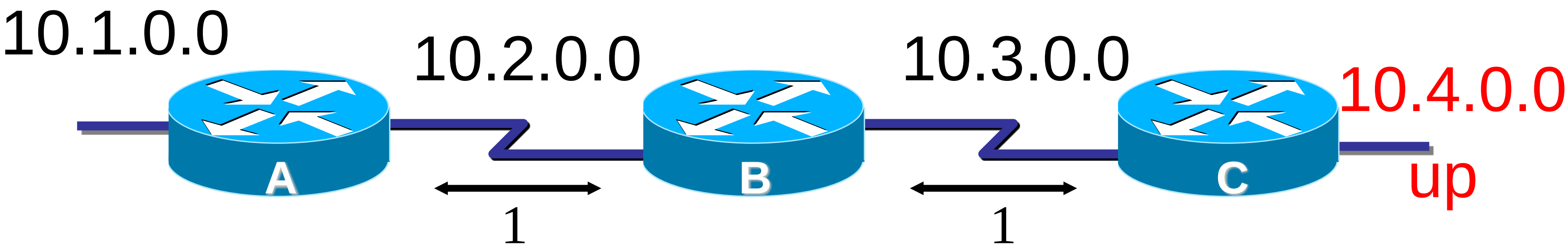
路由算法



Network Protocol	Destination Network	Exit Interface
Connected	10.120.2.0	E0
Learned	172.16.1.0	S0

Routed Protocol: IP

网络发现过程

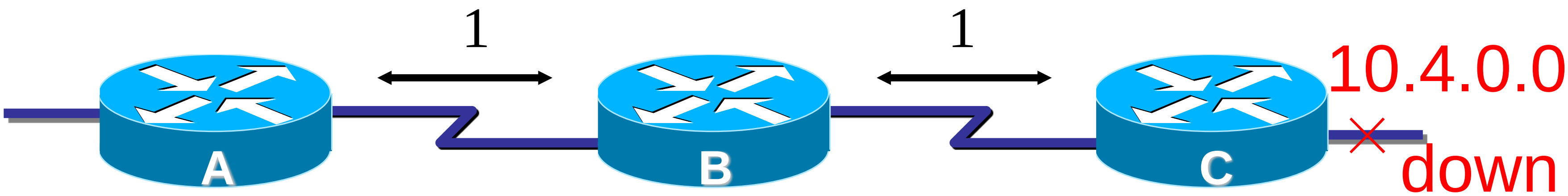


到达信宿10.4.0.0的路由变化

Time	A	B	C	Update
Initialization	∞	∞	∞	信宿不可达
10.4.0.0 up	∞	∞	0	C发现新网络
Step 1	∞	1	0	C→B, 0+1=1
Step 2	2	1	0	B→A, 1+1=2

当如果网络中的最长路径为N,则算法经过N次迭代计算后收敛.即第N步之后,网上的所有路由器都获得到达信宿10.4.0.0的路由信息.

定义距离的最大值

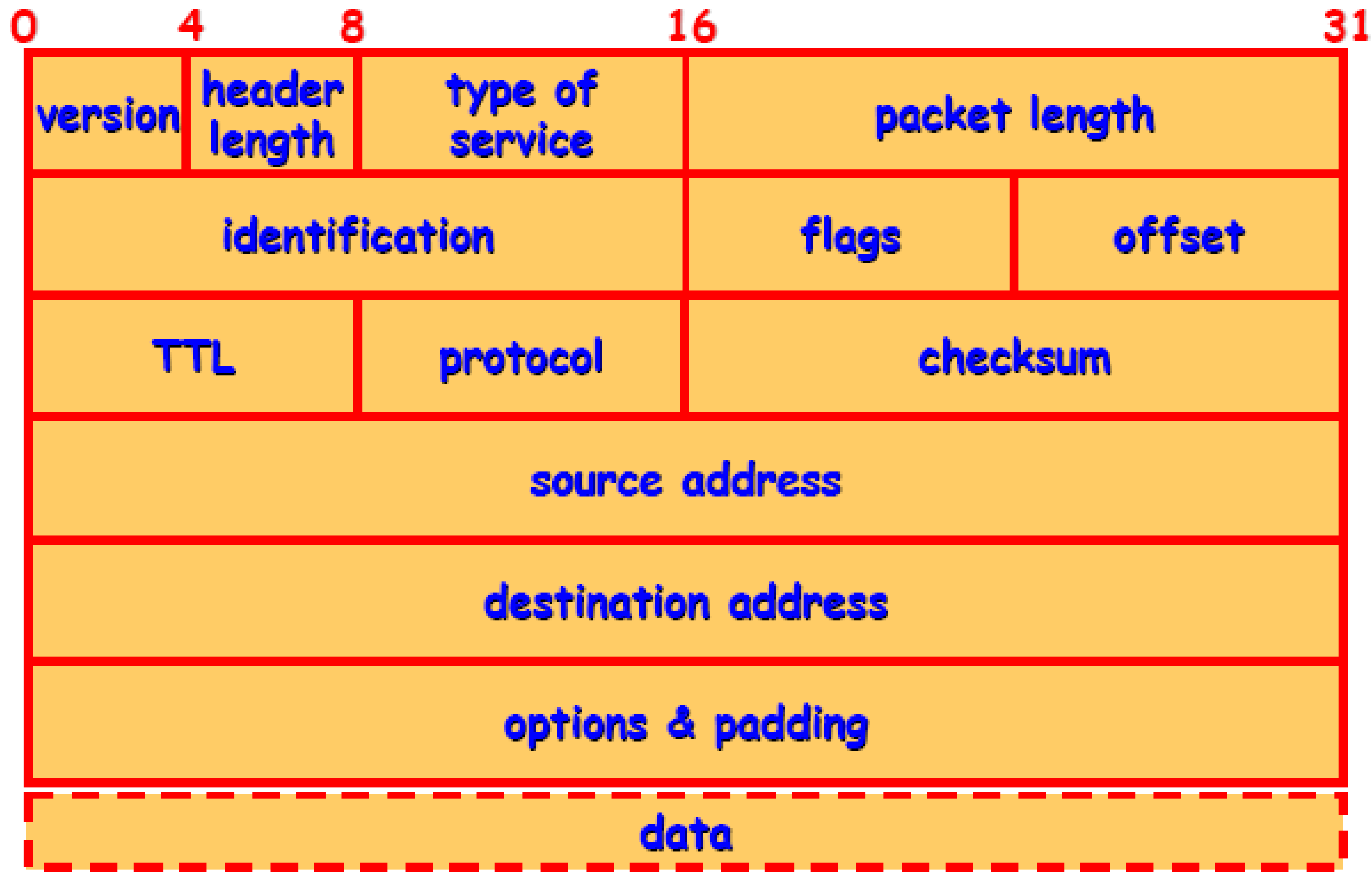


到达信宿40.0.0.0的路由变化(定义Hop最大值为16)

Time	A	B	C	Update
Initialization	2	1	0	信宿可达
10.4.0.0 down	2	1	2	B→C, 1+1=2
Step 1	2	3	2	C→B, 2+1=3
Step 2	4	3	4	B→A, B→C, 3+1=4
Step 3	4	5	4	C→B, 4+1=5
……				
Step 13	14	15	14	C→B, 14+1=15
Step 14	16	15	16	B→A, B→C, 15+1=16
Step 15	不可达	16	不可达	C→B, 15+1=16
Step 16		不可达		discard

收敛!

IP Header: Structure



有限状态机

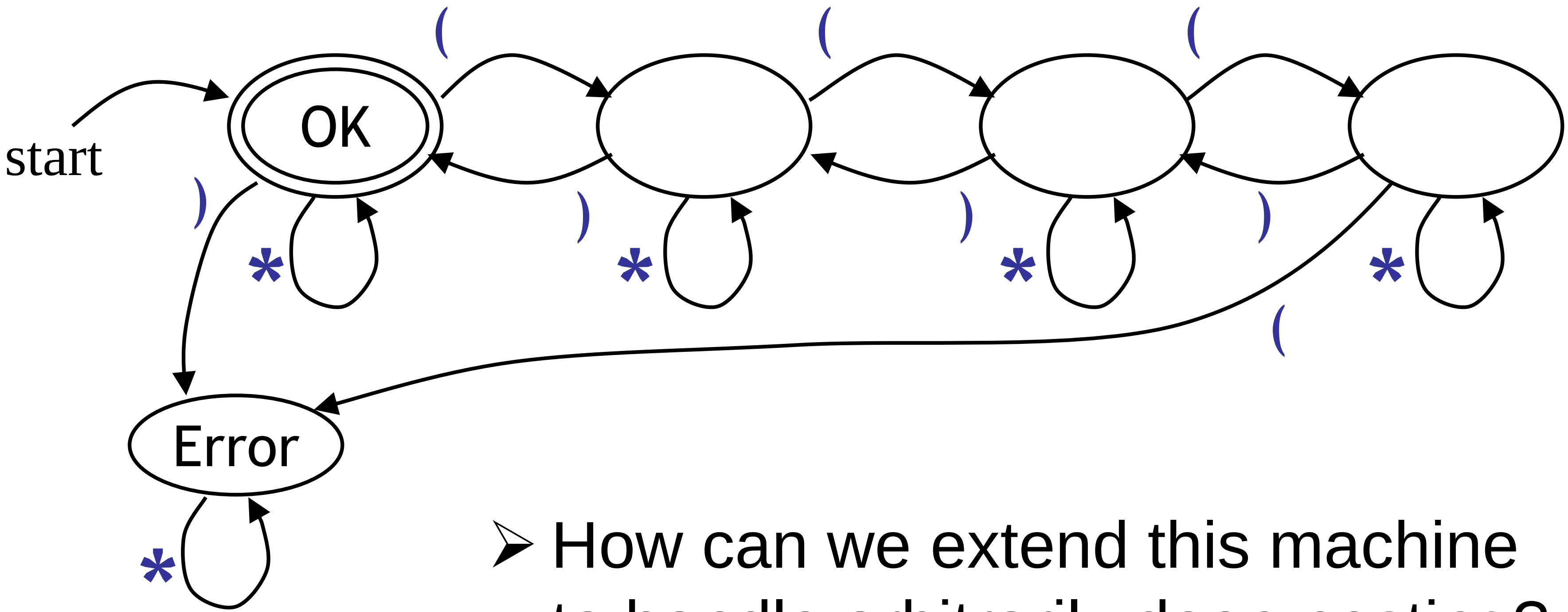
What is a state machine?

- A state machine is a different way of thinking about computation
- A state machine has some number of states, and transitions between those states
- Transitions occur because of inputs
- A "pure" state machine only knows which state it is in—it has no other memory or knowledge
  - This is the kind of state machine you learn about in your math classes
  - When you program a state machine, you don't have that restriction

What is a state machine?

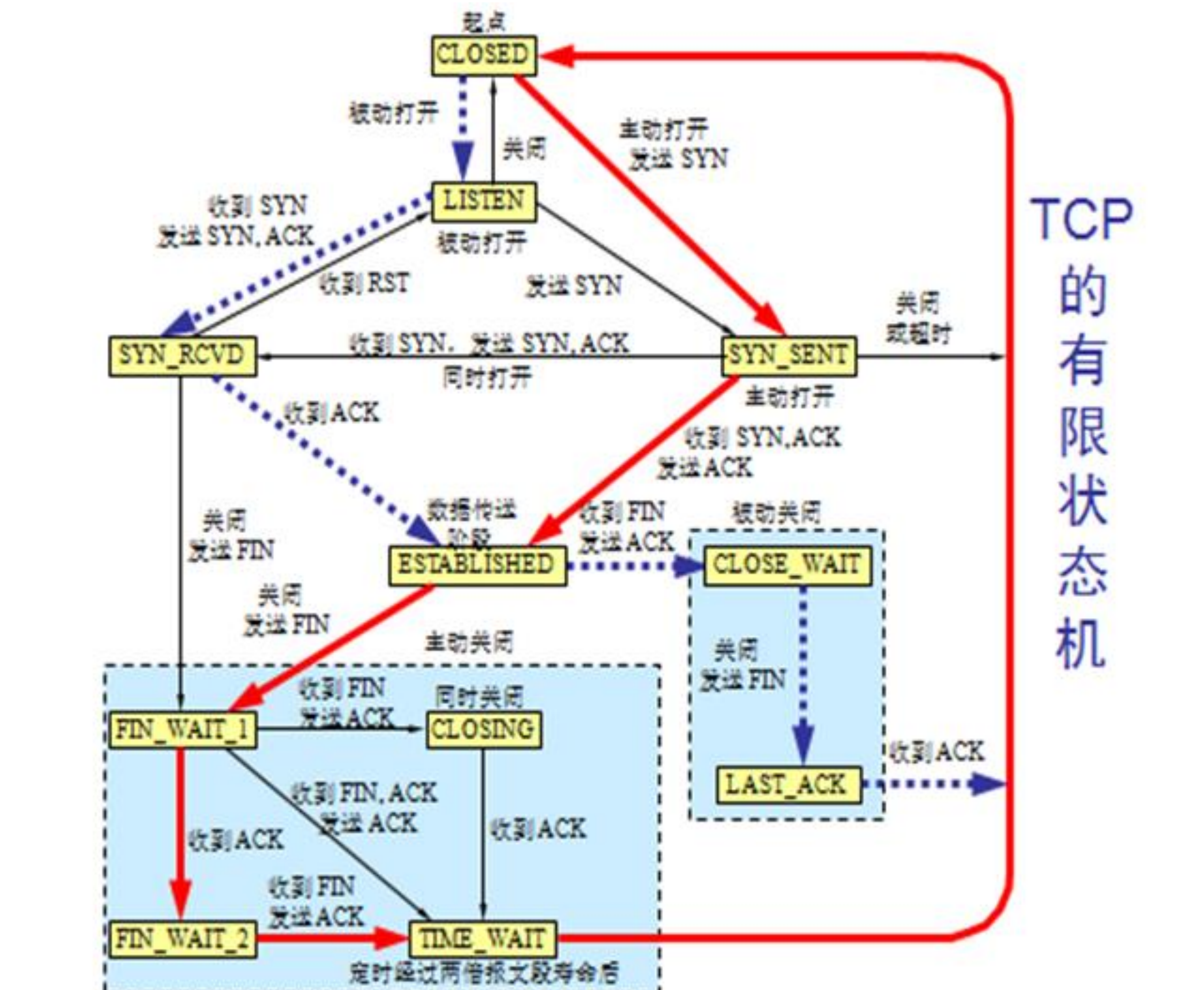
Example Nested parenthesis

- The following example tests whether parentheses are properly nested (up to 3 deep)



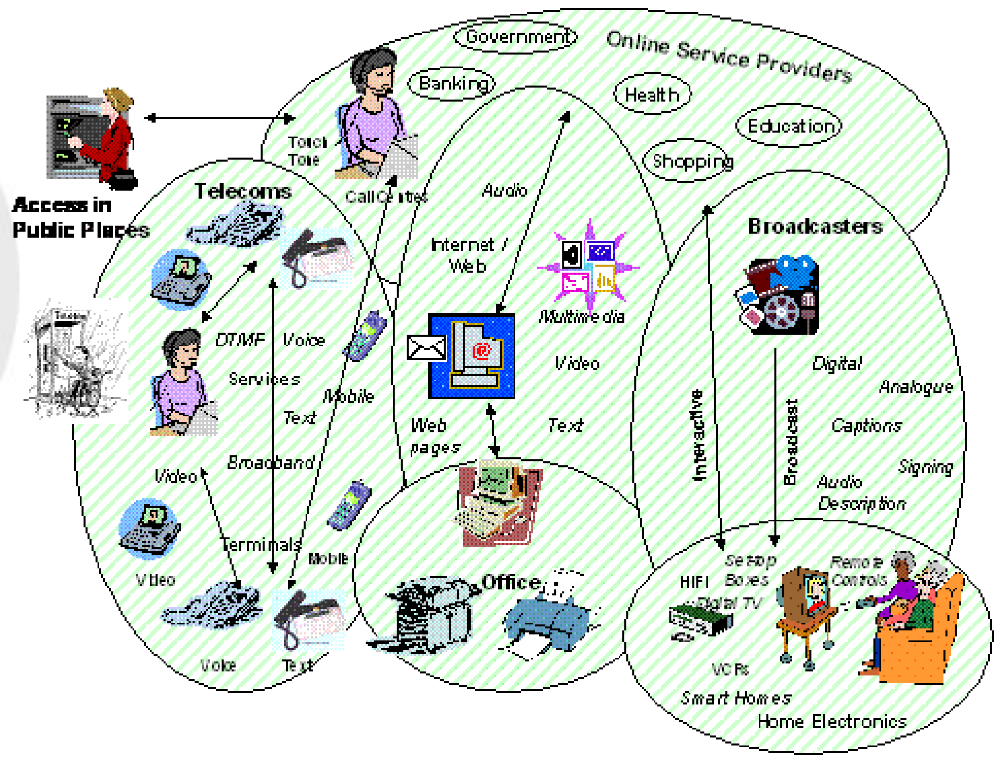
- How can we extend this machine to handle arbitrarily deep nesting?

TCP State machine



网络融合

网络融合 – 技术 vs 业务?

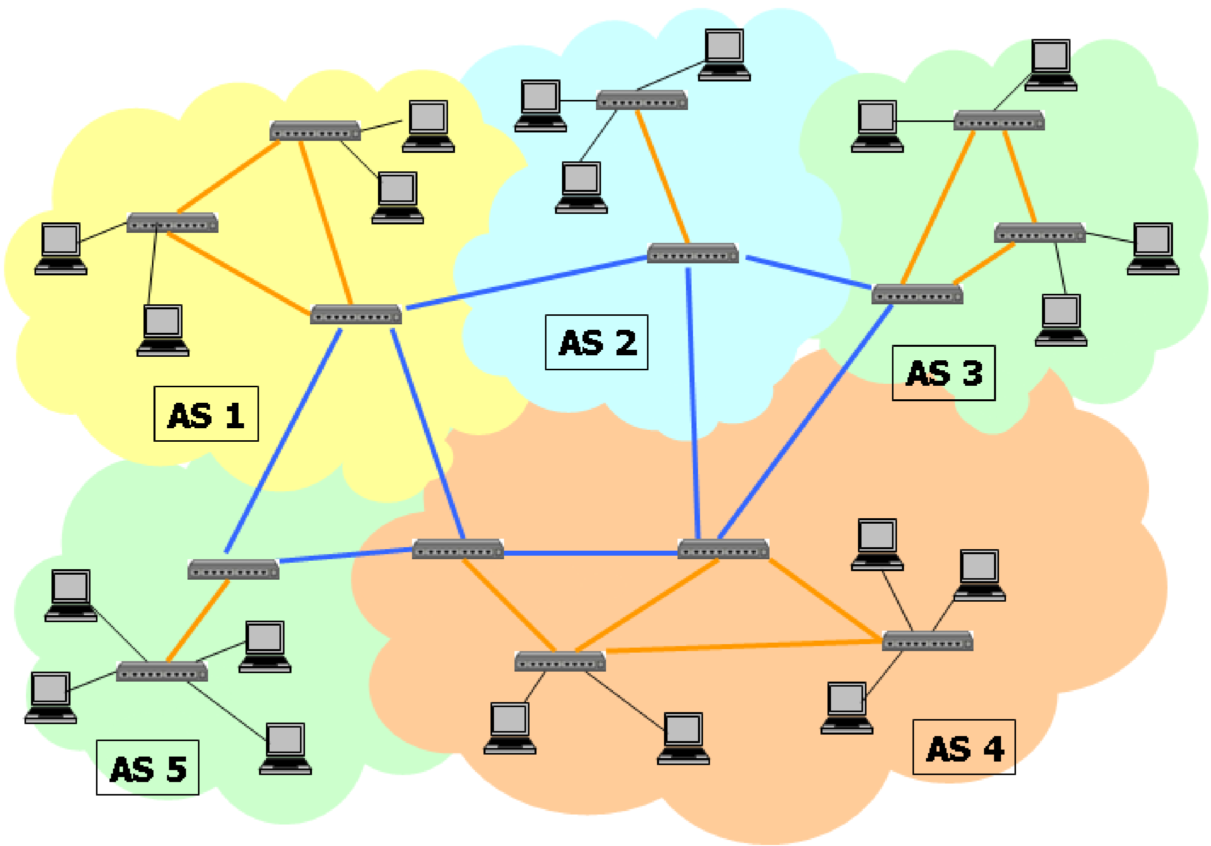

Government
Online Service Providers
Banking
Health
Education
Shopping
Access in Public Places
Touch Tone
Call Centres
Telecoms
Audio
Internet / Web
Multimedia
Broadcasters
DTMF
Voice
Services
Mobile
Text
Video
Web pages
Text
Digital
Analogue
Captions
Signing
Audio Description
Interactive
Broadcast
Broadband
Video
Terminals
Mobile
Video
Voice
Text
Office
HIFI
Set-top Boxes
Digital TV
Remote Controls
VCRs
Smart Homes
Home Electronics

网络融合 – 核心问题

- 早期电信网基于电路交换，时延等QoS有保障
- INTERNET基于TCP/IP协议，支持可扩展性，时延等QoS无保障（例如早期IP电话）
- 如何有效、无缝融合电信网和INTERNET?

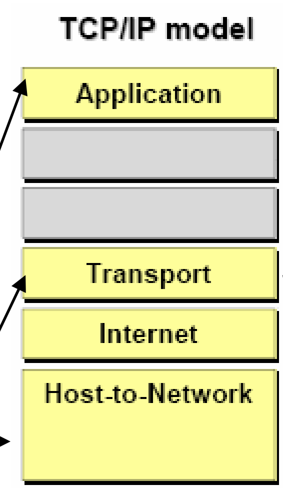
课程小结

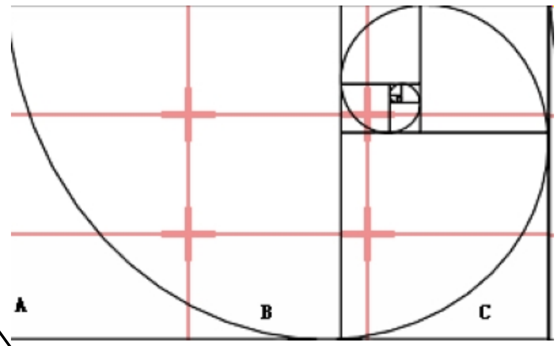
- 信息
- 传输
- 交换



建议

- 打好基本功
- 学习方法论
- 解决实际问题





信号与系统

复变函数

数字通信系统

网络原理

C语言和程序设计

通信与网络

信息论

概率论

数据结构

有限状态机

软件工程

通信信号处理

随机过程

算法原理

形式语言和自动机

排队论

线性代数

数据库

编译原理